

Finite element interface models for the delamination analysis of laminated composites: mechanical and computational issues

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SUMMARY

The finite element analysis of delamination in laminated composites is addressed using interface elements and an interface damage law. The principles of linear elastic fracture mechanics are indirectly used by equating, in the case of single-mode delamination, the area underneath the traction/relative displacement curve to the critical energy release rate of the mode under examination. For mixed-mode delamination an interaction model is used which can fulfil various fracture criteria proposed in the literature. It is then shown that the model can be recast in the framework of a more general damage mechanics theory. Numerical results are presented for the analyses of a double cantilever beam specimen and for a problem involving multiple delamination for which comparisons are made with experimental results. Issues related with the numerical solution of the non-linear problem of the delamination are discussed, such as the influence of the interface strength on the convergence properties and the final results, the optimal choice of the iterative matrix in the predictor and the number of integration points in the interface elements. Copyright © 2001 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Delamination of laminated composite structures has been widely investigated, from both an experimental and a numerical standpoint, because it can often be a cause of a local failure and, sometimes, of a sudden structural collapse.

The procedures usually followed for the numerical simulation of the delamination can be divided into two groups. The first is based on the direct application of fracture mechanics, while the second formulates the problem within the framework of either damage mechanics or softening plasticity, or a coupling between the two. When other material non-linearities can be neglected, linear elastic fracture mechanics (LEFM) provides methods which have been proved quite effective in predicting

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the delamination growth of one or more cracks, provided that their initial position is known in advance. Techniques such as the virtual crack closure (VCC) [1–3], the J -integral [4], the virtual crack extension [5], the stiffness derivative [6], are some of the most used procedures. To model crack propagation, the assumption is made that the delamination propagates when the associated energy release rate is greater than or equal to a critical value, which is indeed a mechanical parameter of the interface, a well-established criterion first introduced by Griffith [7].

For two-dimensional problems the implementation of these methods is sometimes straightforward because the crack front propagates in one dimension. However, for three-dimensional problems, the computational burden increases significantly. Furthermore, they cannot be used to predict the initiation of the delamination and, therefore, they are restricted to problems in which the initial position of the crack, or flow, is known. Even for 2-D applications, problems arise when more than one crack propagate simultaneously [8]. For these reasons, a great effort has been made to develop strategies in which the mechanical behaviour of the interface is modelled on the basis of damage mechanics and/or softening plasticity combined with an indirect introduction of fracture mechanics [9–22]. These procedures, which may be considered to stem from the work of Hilleborg [9], include the ‘cohesive zone model’ [10, 11]. In relation to the simulation of delamination, the method has often been applied in conjunction with interface elements [12–22].

In this paper, the damage law for the interface elements introduced in References [20–22] for general mixed-mode delamination is considered and a modification is proposed in order to ensure that complete delamination occurs simultaneously for all modes, an issue specifically considered also in References [13, 18]. The interface traction relationship is based on a simple bilinear one-dimensional law which is indirectly related to LEFM in that the area of the triangle delimited by the traction/relative displacement curve is equated to the critical energy release rate. A high initial penalty stiffness is assumed to ensure a reasonable pre-crack behaviour.

Special attention must be paid to the choice of the threshold values of the traction components, t_{oi} . It is important for a good prediction of the initiation of the delamination and, in this respect, it should fulfil a failure criterion depending on the material of the interface. Second, it leads to differences between the results obtained by a direct use of LEFM and the results of the interface model. Third, it has a strong influence on the computational efficiency of the finite element simulation, in that the higher it is, the more refined the mesh is required around the delamination front. The last two aspects are discussed in the paper.

It will be shown that the proposed interface model can be derived on the basis of a damage mechanics theory which is similar to the approach adopted for other interface models proposed in the literature [13, 14, 18, 19]. Indeed, the model can be viewed as a regularization of a non-smooth formulation. For a single-mode delamination, the non-smooth model results in an elastic relationship between the relative displacement and the traction at the interface as long as the specific elastic energy stored Ψ is lower than the critical energy release rate G_c , and in a sudden loss of adhesion when $\Psi = G_c$. For mixed-mode delamination, an interaction criterion is adopted which is often used in practical applications [13, 23, 24]. In accordance with this, full delamination occurs, and the crack propagates, as soon as an equation is satisfied which involves the ratios between the dissipated energy for each mode and the critical energy release rate characteristic of the mode itself. Such a criterion has then the advantage that the material parameters involved can be experimentally measured by a series of simple single-mode delamination tests. In the non-smooth model, the specific energy is then dissipated instantaneously and, generally, at one single point of the interface. Therefore, when the penalty stiffness in the pre-crack behaviour tends to infinity, the model reproduces the results of LEFM.

The finite element implementation of the non-smooth formulation would require the use of special crack tip elements as well as very expensive remeshing procedures. Therefore, it turns out to be computationally inefficient with respect to most of the other computational strategies based on LEFM, while sharing with them all the drawbacks mentioned above, such as the difficulties related with the extension to three-dimensional analyses, the prediction of the crack initiation and the application to multiple delamination.

For the regularized model, the energy is dissipated gradually during the delamination process and over a finite area. The results are then expected to be different from the ones provided by the direct application of LEFM. However, it is still possible to consider the total specific energy dissipated at each point of the interface during the delamination process. For a single-mode delamination it is equal to the area under the one-dimensional traction/relative displacement curve and then it is equal to G_c . In the general case of a mixed-mode problem, it will be proved that the interaction criterion proposed in References [23, 24] is still fulfilled if the ratio between the different components of the relative displacement is assumed constant during the damage evolution. As a consequence of this energetic equivalence the numerical results presented show a good-agreement with the direct application of LEFM if sufficiently high values of t_{oi} are chosen. In this case, however, a refined mesh is generally needed around the delamination front in order to avoid spurious oscillations in the structural response.

As far as the solution of the structural problem is concerned, the sharp snap-throughs and snap-backs which often characterize the response of a delaminating composite require sophisticated procedures, or combined static/dynamic solution strategies [25, 26]. Several formulations of the arc-length method are usually adopted, but the optimal choice of strategy is still an open question and here the quadratic arc length, first proposed in References [27, 28] has been used in its modified version presented in References [29, 30].

The influence of the numerical integration rule adopted for the computation of the tangent stiffness matrix and the vector of the residual forces in the interface elements has also been investigated. This issue has been addressed in several papers and the superiority of the Newton–Cotes or the Simpson rule over the traditional Gauss rule is, nowadays, well established [14, 31–35]. In addition, the numerical results presented here point out that increasing the number of integration points, in order to improve the accuracy of the stiffness and of the residual force vector, leads to a less robust solution algorithm and, therefore, to a reduction of the average increment size, whereas the spurious oscillations of the load/displacement curves due to a coarse mesh around the delamination front still remain.

It is also shown that the choice of the tangent matrix at the predictor iteration of each increment plays an important role. A point which undergoes delamination can unload or can be subject to further delamination. Depending on these two alternatives, a secant or a tangent material stiffness, respectively, will be adopted. The second option gives better results in terms of robustness and convergence rate, as will be illustrated for two different examples comparing the rates of convergence of the residual norms for the two different strategies.

2. INTERFACE ELEMENTS FOR THE DELAMINATION ANALYSIS

Let us consider an interface between two parts of a two-dimensional continuum. Assuming that its thickness is thin enough to be considered negligible with respect to the overall geometrical dimensions of the problem, an effective finite element discretization of the interface is obtained by

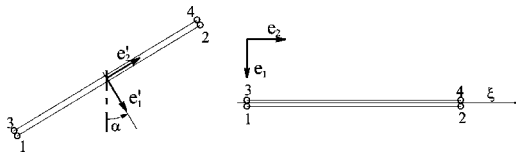


Figure 1. 4-node interface element.

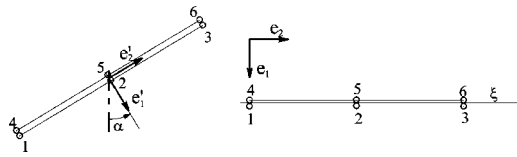


Figure 2. 6-node interface element.

using elements which connect the two parts of the interface whose thickness is taken as exactly zero (before the deformation of the body occurs) so that, in the 2D case under examination, it results in a line.

We will consider linear 4-node (INT4) and quadratic 6-node (INT6) elements, whose master elements are shown in Figures 1 and 2, and a geometrically linear theory. In accordance with the zero thickness, the stresses and the strains are in fact interface tractions and relative displacements. In each element we define a local reference system and each relative displacement is the vectorial sum of a traction/compression mode 1 and a sliding mode 2. At the generic point of the interface, it will be denoted by $\delta = (\delta_1, \delta_2)$ while $\mathbf{t} = (t_1, t_2)$ will indicate the traction.

It is convenient to refer the vector $\mathbf{p} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$ of the displacement parameters, where n is 4 or 6 for the linear and the quadratic elements, respectively, to the global reference system $\{\mathbf{e}_i\}$ and δ to the local element one $\{\mathbf{e}'_i\}$ so that the matrix $\mathbf{B}$ which gives the relationship between δ and $\mathbf{p}$

$$\delta(\xi) = \mathbf{B}(\xi)\mathbf{p} \quad (1)$$

depends upon the rotation of the element and is given by:

$$\mathbf{B}(\xi) = \sum_{i=1}^n \mathbf{B}_i(\xi) \quad \text{with} \quad \mathbf{B}_i(\xi) = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \psi_i(\xi) \quad (2)$$

where, for $i = 1, \dots, n/2$, $\psi_i(\xi)$ is the value at ξ of the shape function related to node i and $\psi_{i+n/2} = -\psi_i$.

2.1. The uncoupled model

In this section two uncoupled relationships will be introduced for the single-mode delamination analysis. Since in mode 1 the different behaviour in compression and in tension must be taken into account to avoid penetration, the following notation will be useful:

$$s_i[x] = \begin{cases} \langle x \rangle & \text{if } i = 1 \\ |x| & \text{if } i = 2 \end{cases} \quad (3)$$

where the symbol $\langle \cdot \rangle$ denotes the Mc Cauley bracket

$$\langle x \rangle = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Furthermore, a pseudo-time parameter is introduced, which is not necessarily the actual time since any dynamic effects are here neglected. Hence, denoting by $[0, T]$ the range of interest, $\tau \in [0, T]$ will be the instant under examination.

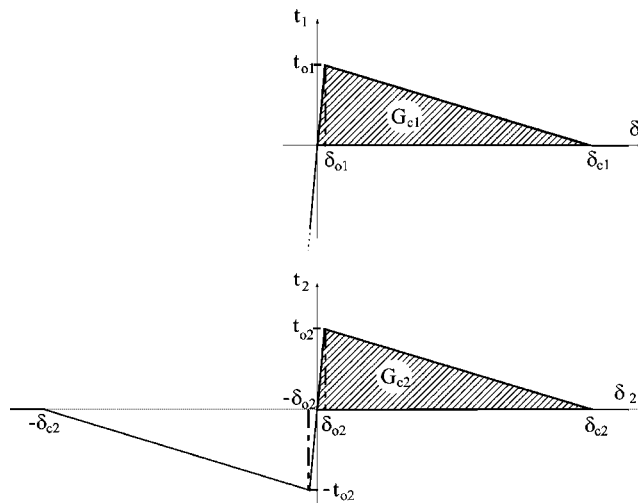


Figure 3. Uncoupled constitutive relationships.

For the uncoupled model the two bilinear, one-dimensional relationships $t_i = t_i(\delta_i)$, which are depicted in Figure 3, are assumed. Their analytical expression is

$$t_i = \begin{cases} K_i \delta_i & \text{if } (s_i[\delta_i])_{\max} \leq \delta_{oi} \\ & \text{or } (i = 1 \text{ and } \delta_i < 0) \\ K_i \delta_i \left[1 - \left(\frac{(s_i[\delta_i])_{\max} - \delta_{oi}}{(s_i[\delta_i])_{\max}} \right) \left(\frac{\delta_{ci}}{\delta_{ci} - \delta_{oi}} \right) \right] & \text{if } \delta_{oi} < (s_i[\delta_i])_{\max} < \delta_{ci} \\ & \text{and } (i \neq 1 \text{ or } \delta_i \geq 0) \\ 0 & \text{if } (s_i[\delta_i])_{\max} \geq \delta_{ci} \\ & \text{and } (i \neq 1 \text{ or } \delta_i \geq 0) \end{cases} \quad (4)$$

where $(s_i[\delta_i])_{\max}$ denotes the maximum value which has been attained by $s_i[\delta_i]$:

$$(s_i[\delta_i])_{\max} = \max_{0 \leq \tau' \leq \tau} s_i[\delta_i(\tau')]$$

and K_i are penalty stiffness parameters.

By inserting $(s_i[\delta_i])_{\max}$ we take into account the irreversibility of the interface damage, so that for the relative displacement history depicted in Figure 4(a) we get the traction response of Figure 4(b). After reaching the value δ^* , for decreasing values of δ an elastic unloading occurs with a reduced stiffness which represents the secant from the current point to the origin. Furthermore, after $(s_i[\delta_i])_{\max}$ has exceeded the critical value δ_{ci} the traction vanishes as a consequence of the complete damage of the interface.

When $i = 1$ and $\delta_i < 0$ relationships (4) ensure no penetration, to within a tolerance which depends upon the penalty stiffness parameter K_1 adopted.

The energy release rate criterion of linear elastic fracture mechanics (LEFM) for the propagation of the delamination in the interface is indirectly used by equating the areas under the traction/relative displacement curves to the critical energy release rates G_{c1} and G_{c2} . Therefore,

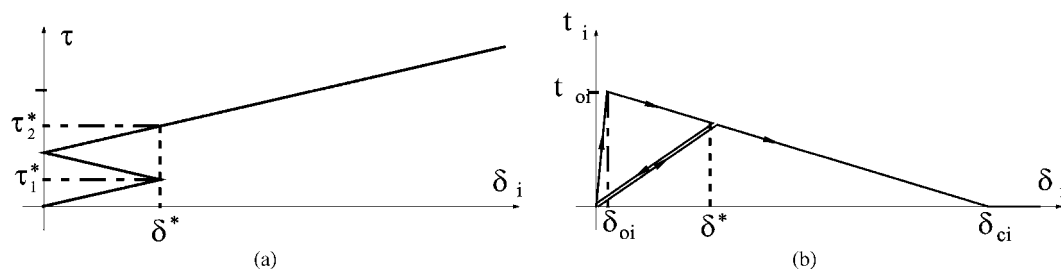


Figure 4. (a) Relative displacement vs time; (b) traction response.

the total energy dissipated during the delamination is correctly computed although at each point it is not released instantaneously as is assumed in LEFM.

It is generally assumed that, for laminated composites, G_{c1} and G_{c2} are characteristic parameters of the interface and do not depend on the whole geometry of the composite nor on the applied loads. They can then be measured by single-mode delamination tests carried out on simple specimens.

The area under the first linear branch of the traction/relative displacement curve for the single-mode i will be denoted with G_{oi} and is given by

$$G_{oi} = \frac{1}{2} K_i \delta_{oi}^2 \quad (5)$$

It represents the specific elastic energy stored, at the point of the interface under examination, just at the beginning of the damaging process (Figure 5).

When the initiation of the delamination is an issue of concern the values t_{oi} should take into account the actual strength of the interface. In this sense, the tractions at the interfaces must be viewed as interlaminar stresses. Currently we use uncoupled values for $t_o = (t_{o1}, t_{o2})$ although they could be chosen in accordance with some interactive failure criterion. The choice of t_o can have a significant influence on the results of the interface model and on its computational efficiency. In particular, as will be shown by the numerical results presented, the higher are the values of t_{o1} and t_{o2} , the more refined is the mesh that is required around the delamination front. Conversely, excessive small values can result in an unsatisfactory prediction of the delamination.

Once the strengths t_{oi} are given, the values δ_{ci} are obtained by the relationship $\delta_{ci} = 2G_{ci}/t_{oi}$. The choice of δ_{oi} is related to the stiffness parameters K_i , which could be computed from the elastic moduli of the interface material and its thickness. However, a more convenient approach is to assume K_i as penalty parameters and then choose δ_{oi} as small as possible within the range which avoids ill-conditioning.

2.2. The mixed model

In most of the two-dimensional engineering applications, delamination involves both an opening mode 1 and a sliding mode 2. The delamination process involves the dissipation of fracture energy in both modes and the total energy release rate G is the sum of the one G_1 associated with the opening mode and the one G_2 associated with the sliding mode

$$G_1 = \int_0^{\delta_1} t_1 d\delta_1 \quad G_2 = \int_0^{\delta_2} t_2 d\delta_2 \quad G = G_1 + G_2 \quad (6)$$



Figure 5. Specific elastic energy at the beginning of delamination.

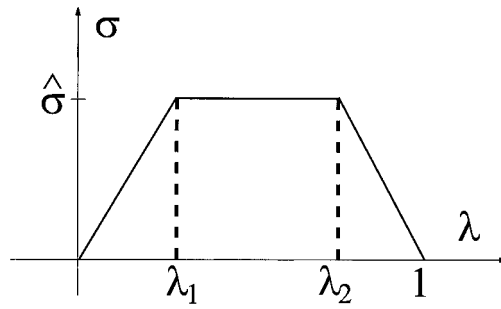


Figure 6. Tvergaard scaled potential.

Some approaches have been proposed to define a single value G_c of the critical energy release rate so as to extend, to the mixed-mode analysis, the criterion $G = G_c$ for the propagation of the crack.

In this line is the widely used Tvergaard model [10] (often referred to as a 'cohesive zone model'), which assumes that the interface law can be derived by a potential $\Phi(\delta_1, \delta_2)$ given by

$$\Phi(\delta_1, \delta_2) = \delta_{c1} \int_0^\lambda \sigma(\lambda') d\lambda' \quad (7)$$

where $\lambda = \sqrt{(\delta_1/\delta_{c1})^2 + (\delta_2/\delta_{c2})^2}$ is a damage parameter which ranges from 0 to 1. For the function $\sigma(\lambda)$ the trapezoidal shape of Figure 6 is sometimes assumed [10, 11] and the total energy dissipated at the end of the damage process turns out to be path independent and is given by

$$G = \frac{1}{2} \hat{\sigma} \delta_{c1} (1 - \lambda_1 + \lambda_2) \quad (8)$$

The formulation has been widely used for the analysis of metals [10, 11]. However, the application of the model to delamination in composite structures is less straightforward basically because, for different mode ratios, full delamination is achieved with quite different values of $G_1 + G_2$. Accordingly, the problem arises of defining a single value of G_c of the critical energy release rate which is strongly dependent on the mode ratio.

In particular, with the fixed-mode ratio (FMR) test, a fixed value G_1/G_2 is defined by the experimental set up and from the experimental results one may obtain G_c appropriate to this particular ratio [36]. However, one may not have experimental results for the particular mode ratio under consideration. In addition, the ratio may be varying.

One particular solution, discussed in Reference [21], is to perform a preliminary computation using LEFM to obtain the mode ratio and then use this value to compute a value of G_c with the aid of an interaction relationship such as the linear relationship given in Reference [23]

$$\left(\frac{G_1}{G_{c1}} \right) + \left(\frac{G_2}{G_{c2}} \right) = 1 \quad (9)$$

or the quadratic relationship considered in Reference [24]:

$$\left(\frac{G_1}{G_{c1}} \right)^2 + \left(\frac{G_2}{G_{c2}} \right)^2 = 1 \quad (10)$$

or, indeed, the generalized ellipse criterion (GEC) of the form [20–22]

$$\left(\frac{G_1}{G_{c1}}\right)^{\alpha/2} + \left(\frac{G_2}{G_{c2}}\right)^{\alpha/2} = 1 \quad (11)$$

where α is a material parameter and will generally assume values between 2 and 4. For $\alpha = 2$ and 4 the linear criterion (9) and the ellipse criterion (10) are recovered, respectively.

Knowing G_{c1} and G_{c2} from appropriate tests in the separate modes, and the mode ratio G_1/G_2 , we may then use (9), (10) or (11) to obtain a G_c such that, at propagation:

$$G_1 + G_2 = G_c \quad (12)$$

This was the approach adopted in Reference [21] for using a fracturing criterion based on (12).

However, we will here consider an alternative approach which requires no ‘pre-solution’ to obtain the mode ratio. It is derived from a constitutive model for the mixed-mode delamination proposed in References [20–22] which fulfils criterion (11) under certain hypotheses on the history of the assigned relative displacements and is based on the introduction of the following parameter $\bar{\gamma}$:

$$\bar{\gamma}(\tau) = \max_{0 \leq \tau' \leq \tau} \gamma(\tau') \quad (13)$$

where

$$\gamma(\tau') = \left[\left(\frac{\langle \delta_1(\tau') \rangle}{\delta_{o1}} \right)^\alpha + \left(\frac{|\delta_2(\tau')|}{\delta_{o2}} \right)^\alpha \right]^{1/\alpha} - 1 \quad (14)$$

and the explicit dependence on τ has been here indicated for the sake of clarity.

The constitutive relationship is then given by

$$\mathbf{t}(\tau) = \begin{cases} \mathbf{K}\delta(\tau) & \text{if } \bar{\gamma}(\tau) \leq 0 \\ [\mathbf{I} - \mathbf{D}(\tau)\mathbf{C}(\tau)]\mathbf{K}\delta(\tau) & \text{if } \bar{\gamma}(\tau) > 0 \end{cases} \quad (15)$$

where $\mathbf{K} = \text{diag}[K_i]$, $\mathbf{I}$ is the identity matrix, and

$$\mathbf{D}(\tau) = \text{diag}[d_i(\tau)] \quad d_i(\tau) = \max \left\{ 1, \frac{\bar{\gamma}(\tau)}{1 + \bar{\gamma}(\tau)} F_{ii} \right\} \quad F_{ii} = \frac{\delta_{ci}}{\delta_{ci} - \delta_{oi}} \quad (16)$$

The matrix $\mathbf{C}(\tau)$ is defined as

$$\mathbf{C}(\tau) = \begin{bmatrix} h[\delta_1(\tau)] & 0 \\ 0 & 1 \end{bmatrix} \quad \text{with } h(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

and ensures that penetration is avoided to within the numerical penalty stiffness adopted.

It is easy to verify that relationships (13)–(16) specialize to Equations (4) for a single-mode delamination problem. Notice also that, as a consequence of the irreversibility of the damage, $\bar{\gamma}$ is a monotonic increasing function of τ . In the sequel, for simplicity, if the explicit dependence of a value upon the pseudo-time parameter is omitted, it is understood that the value is to be computed at the current time τ .

By differentiating relationship (15), and introducing the diagonal matrix $\mathbf{A} = \text{diag}[1/\delta_{oi}^x]$ and the notation:

$$\tilde{\boldsymbol{\delta}} = \begin{bmatrix} \langle \delta_1 \rangle \\ |\delta_2| \end{bmatrix} \quad \tilde{\boldsymbol{\delta}}^x = \begin{bmatrix} \langle \delta_1 \rangle^x \\ |\delta_2|^x \end{bmatrix} \quad \forall x \in \mathbb{R}^+$$

the following expression of the constitutive tangent operator is obtained:

$$\mathbf{K}_t = \begin{cases} \mathbf{K} & \text{if } \bar{\gamma} \leq 0 \\ (\mathbf{I} - \mathbf{DC})\mathbf{K} - \frac{1}{(1 + \bar{\gamma})^{1+\alpha}} \mathbf{FK}(\tilde{\boldsymbol{\delta}} \otimes \tilde{\boldsymbol{\delta}}^{x-1})\mathbf{A} & \text{if } \bar{\gamma} = \gamma > 0 \\ (\mathbf{I} - \mathbf{DC})\mathbf{K} & \text{if } \gamma < \bar{\gamma} > 0 \end{cases} \quad (17)$$

with the further condition:

$$(K_t)_{ij} = 0, \quad j = 1, 2 \quad \text{if } d_i > 1 \quad (18)$$

Notice that the case $\gamma < \bar{\gamma} > 0$ is recovered in the one-dimensional example of Figure 4 when $\tau_1^* < \tau < \tau_2^*$ and then the term $(\mathbf{I} - \mathbf{DC})\mathbf{K}$ represents the secant stiffness characteristic of the unloading path.

Remark 2.1. The proof that relationships (13)–(16) fulfil criterion (11) has been given in the internal report [37] under the assumption of a fixed-mode ratio. For completeness it is reported in Appendix D eliminating the unnecessary hypothesis that $\delta_{o1} = \delta_{o2}$. Later on in the paper the same result will be proved in the general framework of damage mechanics.

2.3. A modified constitutive relationship

In accordance with relationships (13)–(16), complete debonding of one point of the interface is, in general, reached at two-different times for modes 1 and 2. This is inconsistent with the physical evidence that complete debonding allows both the opening and the sliding mode to be activated, simultaneously, with a null value of the resultant traction (unless compression occurs). Some authors add this consistency condition as a further constraint to the evolution equations for the damage parameters [13, 18]. In the model here presented, this problem can be avoided if the ratios δ_{o1}/δ_{c1} and δ_{o2}/δ_{c2} are equal. Since the parameters δ_{o1} and δ_{o2} are mainly related to the penalty stiffness, such an assumption does not give any restrictions and therefore it will be made in the sequel, unless it is differently specified. Introducing the following parameter η :

$$\eta = 1 - \frac{\delta_{o1}}{\delta_{c1}} = 1 - \frac{\delta_{o2}}{\delta_{c2}} \quad (19)$$

the modified constitutive relationship which is here used for the mixed-mode analysis is then as follows:

$$\mathbf{t} = \begin{cases} \mathbf{K}\tilde{\boldsymbol{\delta}} & \text{if } \bar{\gamma} \leq 0 \\ [\mathbf{I} - d\mathbf{C}]\mathbf{K}\tilde{\boldsymbol{\delta}} & \text{if } \bar{\gamma} > 0 \end{cases} \quad (20)$$

where

$$d = \max \left\{ 1, \frac{1}{\eta} \left(\frac{\bar{\gamma}}{1 + \bar{\gamma}} \right) \right\} \quad (21)$$

and the parameter $\bar{\gamma}$ is given in Equation (13).

Remark 2.2. The mixed-mode formulation here presented has close links with that proposed in Reference [13]. The specialization of the formulation is there based on isotropic damage of the interface and can be recast into the following equations:

$$\begin{cases} \mathbf{t} = \mathbf{K}\boldsymbol{\delta} & \text{if } \bar{\psi} \leq 0 \\ \mathbf{t} = (1 - \tilde{\gamma}\bar{\psi})\mathbf{K}\boldsymbol{\delta} & \text{if } 0 < \psi < \frac{1}{\tilde{\gamma}} \end{cases} \quad (22)$$

where $\bar{\psi}$ is a damage parameter defined as:

$$\bar{\psi} = \max_{\tau' \leq \tau} \left\{ \sqrt{\sum_i \left(\frac{\delta_i}{\delta_{oi}} \right)^{2\mu}} - 1 \right\} \quad (23)$$

The meaning of μ is similar to that of α in Equation (14) in that, for a delamination process with a fixed mode ratio, criterion (11) is fulfilled with $\mu = \alpha/2$.

The parameter $\tilde{\gamma}$ is dependent on the value of G_c for each mode. In fact, in order to ensure that, for a case of single-mode delamination, the specific dissipated energy be equal to G_c , the following relationship must be satisfied:

$$\frac{G_{ci}}{G_{oi}} = \frac{\mu}{\mu + 2} \tilde{\gamma} \left[\left(\frac{\tilde{\gamma} + 1}{\tilde{\gamma}} \right)^{(\mu+2)/\mu} - 1 \right] \quad (24)$$

The basic difference between the two models seems to lie in the shape of the traction/relative displacement curve for a single-mode case. With the model proposed in Reference [13] a non-linear law is used so that the non-smoothness of the relationship is reduced. On the other hand, an advantage of the model here presented is that the values of G_{ci} are directly required as input parameters.

3. DAMAGE MECHANICS FORMULATION

In this section the interface model presented in Section 2 is recast in a framework typical of a damage mechanics theory. We first consider the one-dimensional model and then mixed-mode delamination will be addressed.

3.1. One-dimensional model

Let us consider one single-mode and rule out the case of a negative value of the relative displacement, so as to simplify the notation by eliminating any sign issues, and assume the following expression of the free energy potential:

$$\Psi(\delta, d) = \frac{1}{2}(1 - d)K\delta^2 \quad (25)$$

The differentiation of Ψ provides the relationship between the couple (δ, d) and its work-conjugate couple of variables (t, Y) :

$$\begin{aligned} t &= \frac{d\Psi}{d\delta} = (1 - d)K\delta \\ Y &= -\frac{d\Psi}{dd} = \frac{1}{2}K\delta^2 \end{aligned} \quad (26)$$

Remark 3.1. The dual parameters d and Y are internal variables of the model which account for the debonding of the interface and the product $Y\dot{d}$ represents the specific energy dissipating in the damage process at the current time τ .

In this respect it is worth recalling the well-known mechanical interpretation of d as a measure of the microvoids which occur in the interface at a micromechanical scale, that is, referring to an infinitesimal part of the interface, a ratio between the area complementary to the effective load-bearing part and the nominal area [38, 39]. Hence, Y is the specific elastic energy which would be stored if the interface was undamaged, Yd is the specific energy which is actually stored and $Y\dot{d}$ represents the specific energy released when the damage grows, for a fixed value of δ , which also explains the minus sign conventionally adopted in Equation (26)₂.

Equations (26) do not fully define the constitutive behaviour of the interface and an evolution law must be assumed. This can be done by defining a damage threshold Y_c , depending on d , and introducing the following Khun–Tucker conditions:

$$Y \leq Y_c \quad \dot{d} \geq 0 \quad (Y - Y_c) \cdot \dot{d} = 0 \quad (27)$$

with the further constraint

$$0 \leq d \leq 1 \quad (28)$$

The choice of the relationship between Y_c and d results in different damage models. A first possibility consists of assuming that no damage occurs as long as Y keeps constantly lower than the critical value G_c of the energy release rate and that complete damage is obtained as soon as Y attains the value G_c . It can be recovered if the following relationship is assumed between Y_c and d :

$$\begin{aligned} Y_c &= G_c & \text{if } 0 \leq d < 1 \\ Y_c &= \bar{Y} & \text{if } d = 1 \end{aligned} \quad (29)$$

where

$$\bar{Y}(\tau) = \max_{\tau' < \tau} Y(\tau') = \frac{1}{2} K \delta_{\max}^2 \quad (30)$$

Remark 3.2. Notice that (29) is a multivalued law since, for $d = 1$, Y_c can assume infinite values. However, this fact does not have any mechanical implications since, for $d = 1$, $\dot{d}$ is necessarily zero. Then the dissipation $Y\dot{d}$ is zero, as well, so that the value of Y assumes a pure formal meaning.

In accordance with Equation (27)_{2–3}, a positive value of $\dot{d}$ is associated only with a constant value of $Y = Y_c$ or, equivalently, $Y = G_c$. The total specific energy dissipated is then equal to G_c :

$$\int_0^{+\infty} Y \cdot \dot{d} \, dt = G_c \quad (31)$$

Moreover, since this energy can be thought of as being dissipated instantaneously, the difference from the application of LEFM is only that the finite value of the penalty stiffness allows a non-zero elastic relative displacement before the damage of the interface occurs.

The model proposed in Section 2 can be recovered from this non-smooth formulation if the value of the tensile strength is such that the softening lines of the curves in Figure 3 degenerate

to vertical lines or, equivalently, if the values δ_o and δ_c coincide. In that case, the energy G_o introduced in (5), that is the area under the first linear part of the traction/relative displacement curve, coincides with the total area G_c , see also Figure 7.

However, the implementation of such a model in a finite element formulation without any further regularization is extremely cumbersome in terms of CPU time for two reasons. First, the high value of the penalty stiffness, which is recommendable for having a sufficiently accurate pre-crack behaviour, results in an extremely high value of the tensile strength t_o , in accordance with the singularity which is found in the traction profile at a sharp crack tip in a LEFM model. The finite element model should then approximately reproduce such a singularity as well as the high value of the traction gradient immediately before the crack tip. This would require the use of special crack tip elements as well as an adaptive remeshing procedure at each movement of the delamination front, and makes the extension to three-dimensional problems computationally very expensive and conceptually complex. Second, the huge oscillation in the value of the traction would result in severe instabilities of the non-linear solution procedure. Hence, the non-smooth model misses the advantages of the interface model while retaining the drawbacks.

A regularization is then necessary and, as will be proved in Appendix A, the model proposed in Section 2, with $G_c > G_o$, is recovered using the damage mechanics formulation of Equations (26) and (27) if the following regularized relationship between Y_c and d is assumed:

$$\begin{aligned} Y_c &= G_o & \text{if } d = 0 \\ Y_c &= \frac{G_o}{(1 - ((G_c - G_o)/G_c)d)^2} & \text{if } 0 < d < 1 \\ Y_c &= \bar{Y} & \text{if } d = 1 \end{aligned} \quad (32)$$

It is then useful to introduce the regularization parameter η (already defined in (19) referring to the mixed model):

$$\eta = 1 - \frac{G_o}{G_c} = 1 - \frac{\delta_o}{\delta_c} \quad (33)$$

so that relationship (32) can be rewritten as follows:

$$\begin{cases} Y_c = G_o & \text{if } d = 0 \\ Y_c = \frac{(1 - \eta)G_c}{(1 - \eta d)^2} & \text{if } 0 < d < 1 \\ Y_c = \bar{Y} & \text{if } d = 1 \end{cases} \quad (34)$$

In Figure 8 the function defined by Equation (34) is plotted for different values of the parameter η . Clearly the non-smooth model is recovered for $\eta \rightarrow 0$, that is for $G_o \rightarrow G_c$, although in the numerical applications the value of η will be taken very close to 1.

In Appendix A it will be shown that, for $0 < d < 1$, relationship (32), together with (27) and (28), imply that $Y_c = \bar{Y}$. Accordingly, since the dissipation occurs for $0 < d < 1$, that is when $(1 - \eta)G_c \leq \bar{Y} = Y_c \leq G_c/(1 - \eta)$, the total specific energy dissipated in the damage process is given by:

$$\int_0^{+\infty} \bar{Y} \cdot \dot{d} \, dt = \int_0^{+\infty} Y_c \cdot \dot{d} \, dt = \int_{(1-\eta)G_c}^{G_c/(1-\eta)} Y_c \cdot \frac{dd}{dY_c} \, dY_c$$

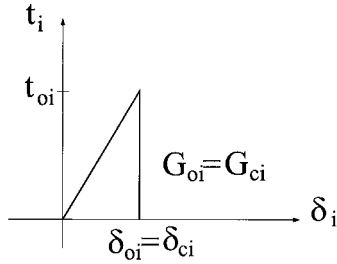


Figure 7. Non-smooth model.

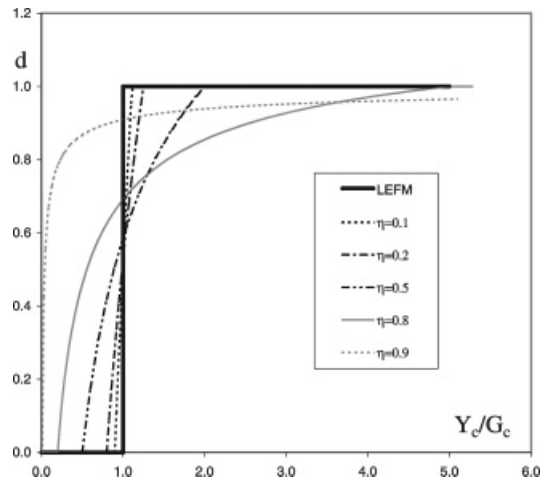


Figure 8. Regularized law.

$$= \int_{(1-\eta)G_c}^{G_c/(1-\eta)} Y_c \frac{1}{\eta} \frac{\sqrt{(1-\eta)G_c}}{2Y_c^{3/2}} dY_c = G_c \quad (35)$$

where, in order to compute dd/dY_c , the inverse function of $(34)_2$ needs to be evaluated.

However, the energy is not dissipated instantaneously and locally at the crack tip, but it is rather dissipated gradually in time and over a finite area of the interface.

Remark 3.3. Notice that relationship (35) implies that the two areas, on the right and on the left, respectively, of the vertical line $d = 1$ in Figure 8 and enclosed between the LEFM and each of the interface model curves, are equal.

3.2. Mixed mode

Let us assume that relationship (19) holds true and choose the following expression of the free energy potential:

$$\Psi(\delta, \mathbf{d}) = \frac{1}{2} [\mathbf{I} - \mathbf{D}(\mathbf{d})\mathbf{C}]\mathbf{K}\delta \cdot \delta \quad (36)$$

where $\mathbf{d} = d(1, 1)$ and $\mathbf{Y} = (Y_1, Y_2)$ is the internal variable which is work-conjugate to $\mathbf{d}$.

Differentiation of relationship (36) provides the relationships

$$\begin{aligned} \mathbf{t} &= d_\delta \Psi = [\mathbf{I} - \mathbf{D}(\mathbf{d})\mathbf{C}]\mathbf{K}\delta \\ \mathbf{Y} &= d_{\mathbf{d}} \Psi = \begin{bmatrix} \frac{1}{2} h(\delta_1) K_1 \delta_1^2 \\ \frac{1}{2} K_2 \delta_2^2 \end{bmatrix} \end{aligned} \quad (37)$$

Introducing the following convex function:

$$f(\mathbf{Y}) = \left(\frac{Y_1}{G_{c1}} \right)^{\alpha/2} + \left(\frac{Y_2}{G_{c2}} \right)^{\alpha/2} \quad (38)$$

one can define the damage domain $\mathcal{D}$

$$\mathcal{D} = \{\mathbf{Y} : f(\mathbf{Y}) \leq \omega_c\} \quad (39)$$

which depends on the non-dimensional threshold value ω_c . The following evolution law for the internal variables $\mathbf{d}$ and $\mathbf{Y}$ is then introduced:

$$\dot{\mathbf{d}} = \dot{d} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (40)$$

with the further conditions

$$f(\mathbf{Y}) \leq \omega_c \quad \dot{d} \geq 0 \quad [f(\mathbf{Y}) - \omega_c] \dot{d} = 0 \quad (41)$$

and the additional constraint (28).

A 'non-smooth' model is then introduced by assigning the following relationship between ω_c and d , analogous to (29):

$$\begin{aligned} \omega_c &= 1 & \text{if } 0 \leq d < 1 \\ \omega_c &= \bar{\omega} & \text{if } d = 1 \end{aligned} \quad (42)$$

where

$$\bar{\omega} = \max_{\tau' \leq \tau} f[\mathbf{Y}(\tau')] \quad (43)$$

Hence, for the non-smooth model, no damage occurs as long as $f(\mathbf{Y})$ keeps lower than 1, that is as long as $\mathbf{Y}$ is strictly inside $\mathcal{D}$, whereas complete damage is obtained when $f(\mathbf{Y})$ attains the value 1. As for the (42)₂, analogous considerations as in Remark 3.2 hold.

Although it would be theoretically possible that, during the damage process in which d increases from 0 to 1, Y_1 and Y_2 change while keeping $f(\mathbf{Y}) = 1$, it is reasonable to rule out such a case and assume that the whole damage occurs for fixed values of Y_1 and Y_2 . Accordingly, the following relationship holds:

$$\left(\frac{\int_0^{+\infty} Y_1 \cdot \dot{d}_1 dt}{G_{c1}} \right)^{\alpha/2} + \left(\frac{\int_0^{+\infty} Y_2 \cdot \dot{d}_2 dt}{G_{c2}} \right)^{\alpha/2} = 1 \quad (44)$$

Also for the mixed-mode formulations, Y_1 and Y_2 can then be regarded as the energy release rates related with modes 1 and 2, respectively, and in the limit case, when the penalty stiffness goes to infinity, we recover the LEFM formulation.

Notice also that relationships (41) do not fulfil a normality law to the convex damage domain $\mathcal{D}$ since the normal to $\mathcal{D}$ at a point $\mathbf{Y}$ of its boundary is equal to

$$\mathbf{n}(\mathbf{Y}) = \frac{\alpha}{2} \begin{bmatrix} \frac{1}{Y_1} \left(\frac{Y_1}{G_{c1}} \right)^{\alpha/2} \\ \frac{1}{Y_2} \left(\frac{Y_2}{G_{c2}} \right)^{\alpha/2} \end{bmatrix} \quad (45)$$

whereas $\dot{\mathbf{d}}$ is collinear with the vector $(1, 1)$.

The same arguments as for the one-dimensional case justify the necessity of introducing a regularized model. In fact, as will be shown in Appendix B, the mixed-mode formulation presented in Section 2.2.1 can be regarded as a regularization of the non-smooth damage model above illustrated and is recovered assuming the following relationship between ω_c and d :

$$\begin{cases} \omega_c = (1 - \eta)^{\alpha/2} & \text{if } d = 0 \\ \omega_c = \left[\frac{1 - \eta}{(1 - \eta d)^2} \right]^{\alpha/2} & \text{if } 0 < d < 1 \\ \omega_c = \bar{\omega} & \text{if } d = 1 \end{cases} \quad (46)$$

where again η is the regularization parameter of Equation (19), which can be rewritten as follows:

$$\eta = 1 - \frac{G_{o1}}{G_{c1}} = 1 - \frac{G_{o2}}{G_{c2}} \quad (47)$$

The damage domain $\mathcal{D}$ then changes during the damage process, since the threshold ω_c has the initial value $(1 - \eta)^{\alpha/2}$, approaches the value $(1 - \eta)^{-\alpha/2}$ when $d \rightarrow 1$, and can indefinitely increase when $d = 1$, together with $\bar{\omega}$.

Also the regularized model does not fulfil a normality law since the normal to $\mathcal{D}$ is, in general, not collinear with the vector $(1, 1)$.

Remark 3.4. It is important to point out that the proposed interface model is consistent with the second principle of thermodynamics because the Clausius–Duhem inequality is fulfilled. Actually, since variations of temperature are not considered in the present context, the Clausius–Duhem relationship results in the following expression:

$$\mathbf{t} \cdot \dot{\boldsymbol{\delta}} - \dot{\Psi} = \mathbf{Y} \cdot \dot{\mathbf{d}} \geq 0, \quad (48)$$

which holds true because $\mathbf{Y} \geq \mathbf{0}$ and $\dot{\mathbf{d}} \geq \mathbf{0}$, componentwise.

As far as the total specific energy dissipated is concerned, it is dependent on the assigned history of the relative displacement. However, if a fixed mode ratio is assumed so that $\delta_2 = z\delta_1$, it will be proved in Appendix C that the generalised ellipse criterion (44) is fulfilled.

4. NUMERICAL SOLUTION

The softening behaviour associated with the damage of the interface often results in a global structural response which, in terms of load–displacement curves, is characterized by snap-throughs and snap-backs. These are often related to a limit point, that is a singularity of the tangent stiffness

of the finite element model. Traditional incremental procedures generally fail to overcome these limit points so that more sophisticated strategies need to be adopted such as the arc length methods.

Starting from a pioneering work of Riks [40], who first proposed to add to the structural problem an additional constraint which enforces, at each increment, that a combined norm of the changes of the displacements and of the load (arc length) be equal to a given value, several authors [20, 21, 41–43] developed different versions of the arc length method in order to handle different types of non-linear analyses.

In particular, problems such as the delamination of composite laminates have the peculiarity that the non-linear constitutive behaviour can be limited to the small region of the interface where the crack propagates, while the remaining part of the structure is often modelled as linearly elastic. Even when further non-linearities are introduced in the finite element model, as when geometric non-linearity is taken into account or when some damage-plasticity laws for the composite layers are assumed, the presence of limit points in the structural response is still mainly related with the crack propagation at the interface.

This consideration suggested [13, 15, 18] that attention should be focused on those arc length methods in which the control parameter, which is taken as fixed in the increment, is a function of a limited set of degrees of freedom (local control), namely the ones which are directly related to the non-linear behaviour, in this case the relative displacements in the interface elements at the delamination front.

Accordingly, this approach was initially followed by adopting the local control procedure illustrated in Reference [44], a method which is close to that proposed in References [13, 18]. However, some preliminary numerical results, which have been presented in Reference [45], indicated that the method was less robust than the quadratical arc length [27, 28], in its modified form recently presented in References [29, 30]. The latter method has therefore been used for the current numerical applications. Nevertheless, the possibility of improving the numerical stability and the convergence rate via a local control procedure needs further investigation and will be the subject of future work.

Referring to the above-quoted references, for major details, let us briefly recall that in the k th iteration, within a generic increment of the full Newton method, the following equation is solved:

$$\hat{\mathbf{K}}_t^k \delta \mathbf{p}^k = \lambda^{k+1} \mathbf{q}^e - \mathbf{q}^i(\mathbf{p}^k) = \mathbf{g}^k + \delta \lambda^k \mathbf{q}^e, \quad k = 0, 1, \dots$$

where $\hat{\mathbf{K}}_t^k$ is the structural stiffness matrix, $\delta \mathbf{p}^k$ is the change of the displacements in the iteration, $\mathbf{q}^i(\mathbf{p}^k)$ is the vector of the internal forces, $\lambda^{k+1} = \lambda^k + \delta \lambda^k$ is the load multiplier, at the end of the iteration, of the external force vector $\mathbf{q}^e$ (proportional loading is assumed), and $\mathbf{g}^k = \lambda^k \mathbf{q}^e - \mathbf{q}^i(\mathbf{p}^k)$ is the residual force vector at the beginning of the k th iteration.

Accordingly, the superscript 0 will denote the values obtained after reaching convergence at the previous increment, and the total change $\Delta \mathbf{p}^{k+1} = \mathbf{p}^{k+1} - \mathbf{p}^0$ is written as follows:

$$\Delta \mathbf{p}^{k+1} = \Delta \mathbf{p}^k + \delta \bar{\mathbf{p}}^k + \delta \lambda^k \delta \mathbf{p}_t^k \quad (49)$$

where the vectors $\delta \bar{\mathbf{p}}^k = [\hat{\mathbf{K}}_t^k]^{-1} \mathbf{g}^k$ and $\delta \mathbf{p}_t^k = [\hat{\mathbf{K}}_t^k]^{-1} \mathbf{q}^e$ are usually referred to as the iterative and tangential displacements, respectively, while $\delta \lambda^k = \lambda^{k+1} - \lambda^k$ is the unknown change of the load factor.

In the quadratic arc length the following equation is added as an additional constraint:

$$\|\Delta \mathbf{p}^{k+1}\| = \Delta l \quad (50)$$

The arc length Δl is assumed as fixed in the increment whereas an automatic incrementation procedure is used to update its value at increment $i + 1$, known the value at increment i , via the formula

$$\Delta l_{i+1} = \sqrt{\frac{I_d}{I_p}} \Delta l_i \quad (51)$$

where I_p is the number of iterations required for getting convergence at increment i and I_d is a user defined number of ‘desired’ iterations.

Two different issues arise at the first ‘predictor’ iteration ($k = 0$) and during the next, possible, ‘corrector’ ones ($k \geq 1$). At the first iteration, Equations (49) and (50) yield

$$\Delta \lambda^o = \lambda^1 - \lambda^o = \pm \frac{\Delta l}{\|\delta \mathbf{p}_t^0\|} \quad (52)$$

Different methods have been proposed for choosing the sign in Equation (52), see References [28, 30]. In the numerical applications carried out, no problems of bifurcation have been encountered and then the same sign has been taken as that of the determinant of the stiffness matrix, which can approximately be evaluated by considering the number of negative pivots resulting from the factorization of $\hat{\mathbf{K}}_t$.

At the corrector iterations Equation (50) results in a quadratic equation in $\delta \lambda^k$ which, in general, will give two real or imaginary roots $\delta \lambda_k^1$ and $\delta \lambda_k^2$. The latter case, which has rarely been experienced during the numerical applications for the interface model, is a warning for a too large increment size Δl . The analysis must then be restarted from the last converged increment but with a reduced arc length.

In the usual case when the roots are real, two different values $\mathbf{p}_1^{k+1}(\delta \lambda_1^k)$ and $\mathbf{p}_2^{k+1}(\delta \lambda_2^k)$ of the displacement vector are associated with them:

$$\mathbf{p}_j^{k+1}(\delta \lambda_j^k) = \mathbf{p}^0 + \Delta \mathbf{p}^k + \delta \bar{\mathbf{p}}^k + \delta \lambda_j^k \delta \mathbf{p}_t^k, \quad j = 1, 2 \quad (53)$$

For problems characterized by relatively smooth limit points, the ‘angle-criterion’ is generally adopted. With this approach, one computes the two angles between the new incremental changes of $\mathbf{p}$, $\Delta \mathbf{p}_j^{k+1} = \mathbf{p}_j^{k+1}(\delta \lambda_j^k) - \mathbf{p}^0$, and the incremental change at the beginning of the iteration, $\Delta \mathbf{p}^k$. The value of $\delta \lambda_j^k$ which corresponds to the minimum of such angles is then chosen.

However, in the examples presented in this paper, it has proved fundamental to follow the procedure recently proposed in Reference [29], see also Reference [30], that is to choose the root which provides the minimum residual norm $\|\mathbf{g}^{k+1}\|$. In fact, this method is necessary when sharp snap-backs are encountered in the structural response. As will be shown by several numerical examples, this is the case of the delamination analysis where sharp snap-backs can be either a real characteristic of the structural response or the result of spurious oscillation related to an insufficient mesh refinement around the crack front.

The key idea underneath the criterion for choosing the root is illustrated in Figure 9. Strictly, it refers to a one-dimensional case and to the so-called ‘spherical’ arc length method [28], in which the change of λ is also added in the constraint equation, since in that case the concept can graphically be explained more easily.

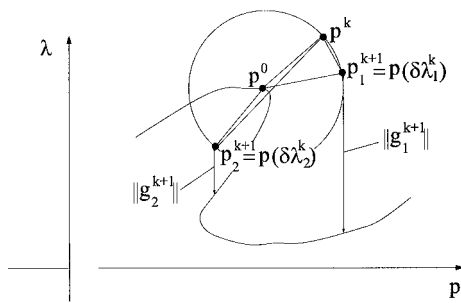


Figure 9. Minimum residual criterion for the choice of the root.

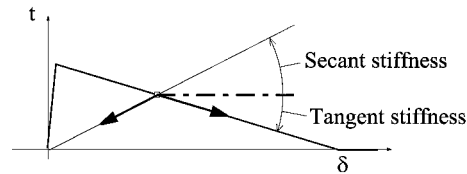


Figure 10. Material stiffness at the predictor.

4.1. Iterative matrix at the predictor

Let us consider a point of the interface at which, during the generic increment i , the delamination process is developing so that the last converged value of the damage parameter d , d_i^c , is greater than the one, d_i^0 , at the beginning of the increment, being $0 < d_i^c < 1$. In this case, in terms of the damage parameter $\bar{\gamma}$, at the beginning of the next increment, $i + 1$, it turns out to be $\gamma_{i+1}^0 = \bar{\gamma}_{i+1}^0$ and then, in accordance with Equation (17), the tangent constitutive operator is used to form the stiffness matrix, computed via (17)₂.

This is not straightforward because it is not known in advance whether at that point, during the new increment, $i + 1$, unloading or further delamination is going to occur. In the latter case expression (17)₂ gives the best predictor material stiffness, whereas in the former case the secant material stiffness provided by expression (17)₃ is more appropriate, see also Figure 10, which refers to a one-dimensional case.

A similar issue appears in other contexts, such as in elastoplasticity when, at the beginning of an increment, the stress at the Gauss point under examination relies on the boundary of the elastic domain. In this latter case some analysts recommend that an elastic behaviour is to be *a priori* assumed, leaving the use of the tangent operator for the next corrector iterations, if necessary. Although it seems to be theoretically impossible to find out a general optimal strategy for addressing an issue which is problem dependent, in Reference [46] it has been shown that, for a general class of elastoplastic problems, the choice of the tangent operator at the predictor iteration results in enhanced convergence properties of the solution algorithm.

As will be detailed later, a similar result has also been obtained in the present context for the analysis of delamination and the numerical results presented show that not only is the convergence rate improved if the tangent material stiffness at the predictor is assumed instead of the secant one but, more important, the robustness of the solution algorithm is significantly enhanced.

Although a deeper investigation for cases like cyclic load histories is probably needed to confirm the general validity of this last statement, it appears that a point which is currently undergoing delamination during one increment is likely to be subject to further damage in the next one. Furthermore, the increased robustness, which is even more significant than the improvement of the convergence rate, is probably related with the strategy adopted for overcoming the limit points,

since the choice of a secant or of a tangent constitutive operator often results in a positive or negative determinant of the structural stiffness matrix and, therefore, in a different sign of the predictor change of the load factor in Equation (52).

4.2. Numerical integration in the interface elements

A peculiarity of the interface model is the fact that the Gauss rule is not recommended for the numerical integration of the tangent stiffness matrix and the internal force vector of the interface elements [14, 31–35].

This issue has been specifically discussed in Reference [31], where an explanation has been provided for some numerical results presented in some papers, mainly related to the use of interface elements for the analysis of geomechanics or concrete problems [32–34], which show a superior performance of the Newton–Cotes integration rule with respect to the Gauss quadrature commonly adopted for the 2D and 3D continuum elements. In Reference [31] it has been shown that, for linear elements, the application of Gauss quadrature results in a coupling between the degrees of freedom of different node sets and then in an oscillation of the traction profile, for high values of the traction gradients, which is not recovered if a Newton–Cotes integration rule is used. Similar results are also found for quadratic elements although in that case some more detailed considerations need to be made.

Furthermore, because of the non-smooth profile of the stress in an element which is only partially delaminated, a high number of integration points might yield a more accurate estimation of the stiffness and of the residual forces [14, 35]. However, some numerical results that will be presented here suggest that the increased stress accuracy does not necessarily allow the use of a coarser mesh and indeed the overall robustness of the solution algorithm can be negatively influenced. As a result, if the mesh is not sufficiently refined around the delamination front the spurious jumps of the load/displacement curves still remain but, in addition, the average increment size evaluated by the automatic incrementation procedure is considerably reduced.

5. NUMERICAL RESULTS

The interface model described in the previous sections has been implemented in a research version of the finite element code LUSAS [47].

In this section some numerical results will be presented and discussed for two different analyses of mode I and mixed-mode delamination. A comparison is also made with the results obtained using the virtual crack closure method [1–3], that is a direct application of LEFM, and with available experimental data [8, 48].

5.1. Numerical simulation of a DCB test

As a pure mode I problem a double cantilever beam has been analysed under displacement control. The geometry is described in Figure 11 and the material data are given in Table I. Strictly, a 3D analysis would be needed since the delamination front is not precisely a straight line. Indeed in earlier work by the second author [22] a three-dimensional analysis was applied. The results indicated that the 2D plane strain formulation gave a very reasonable approximation.

First, a regular mesh of 4×400 eight-node plane strain (Q8) elements and 280 six-node interface (INT6) elements has been adopted. In Figure 12 the vertical reaction at point P has been plotted

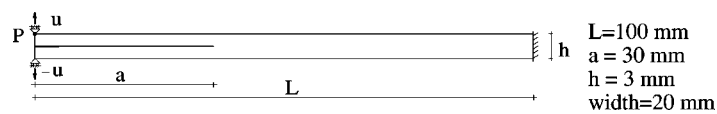
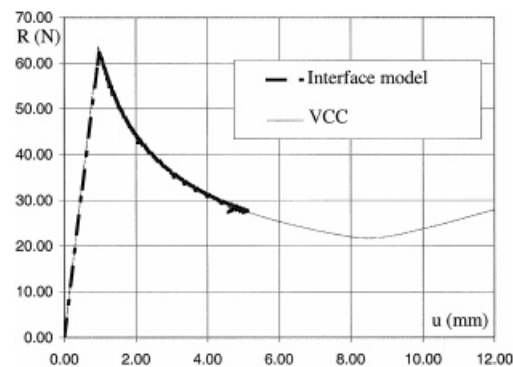


Figure 11. DCB: geometry.

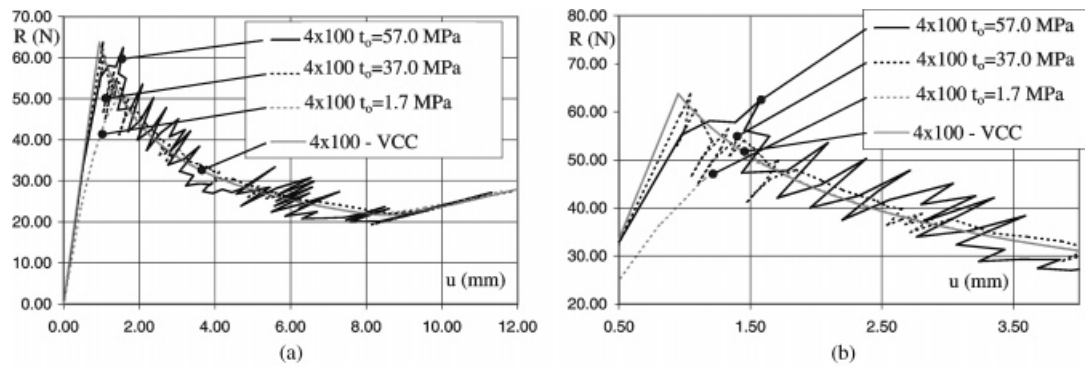
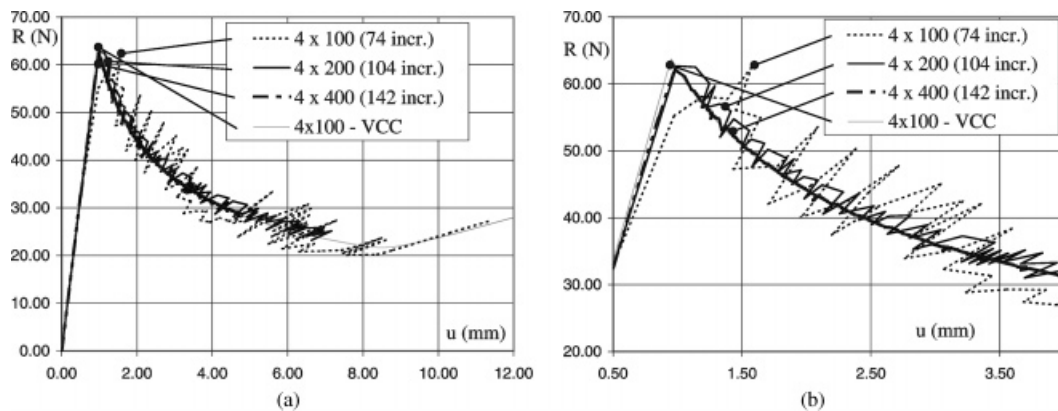
Table I. DCB specimen: material data.

Composite				
$E_{11}(\text{GPa})$	$E_{22} = E_{33}(\text{GPa})$	$G_{12}(\text{GPa})$	$\nu_{12} = \nu_{13}$	ν_{23}
135.3	9.0	5.2	0.24	0.46
Interface				
$G_c(\text{N mm}^{-1})$		δ_o	$t_o(\text{MPa})$	
0.28		10^{-7}	57.0	

Figure 12. DCB: reaction at P vs prescribed displacement.

against the vertical displacement u for the interface model and the VCC method. It is worth noting that the load–displacement curves for DCB analyses presented in the literature often refer to the relative displacement between the two points on the left of Figure 11 where the displacements are prescribed. The latter would be equal to $2u$.

In Figures 13 the results of a sensitivity analysis to the tensile strength have been presented. The spurious oscillations show that, for a value of $t_o = 57 \text{ MPa}$, the mesh used is too coarse and Figures 14 show the improvements obtained by refining the mesh. Notice, however, the increase of the necessary number of increments. It can be seen that the solution obtained with the low tensile strength ($t_o = 1.7 \text{ MPa}$) is in good agreement with the VCC solution for the late propagation phase but the solution is much too flexible in the early loading stages. The later example with mixed-mode fracture will show that this early discrepancy can, for some problems, lead to misleading results even in the later loading stages.

Figure 13. (a) DCB: different t_o ; (b) DCB: different t_o (zoom).Figure 14. (a) DCB: different meshes, $t_o = 57.0$ MPa; (b) DCB: different meshes, $t_o = 57.0$ MPa (zoom).

Next, the influence of the material stiffness used at the predictor iteration is studied. The two methods discussed in Section 4.1 will be referred to as ‘tangent predictor’ and ‘secant predictor’ procedures.

Figure 15(a) refers to the mesh with 4×100 Q8 elements, with 70 six-node interface elements, and to a tensile strength of $t_o = 1.7$ MPa. The response is relatively smooth since it is characterized by one snap-through and no snap-backs.

In this case both procedures are robust but the superiority of the tangent predictor is testified by the improved rate of convergence which results in a reduced number of iterations with respect to the one required for the secant predictor method.

In fact, in accordance with the automatic incrementation procedure adopted, see Equation (51), the improved convergence also yields an increase of the average increment size and then of the arc length Δl .

A more important feature of the tangent predictor procedure is its satisfactory robustness even when very sharp snap-throughs and snap-backs are encountered. The curves in Figure 15(b) have

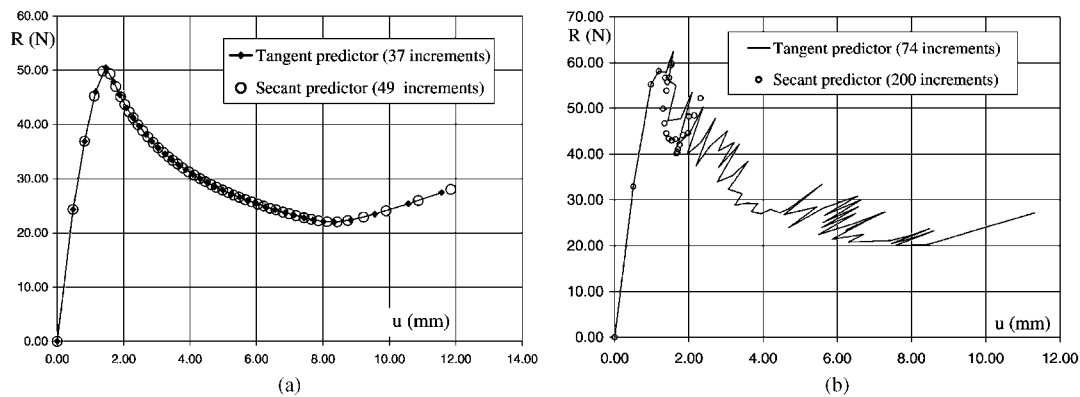


Figure 15. (a) Response with $t_0 = 1.7$ MPa; (b) Response with $t_0 = 57.0$ MPa.

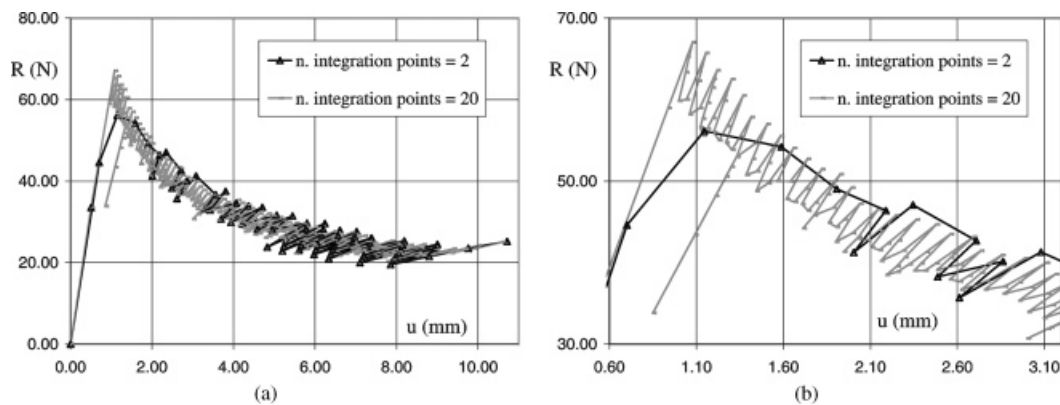


Figure 16. (a) Different number of integration points; (b) different number of integration points (zoom).

been obtained with $t_o = 57.0$ MPa. It shows a case in which several snap-backs are due to a coarse mesh rather than to the real structural behaviour.

One may argue that the problem of having sharp snap-backs can be solved by refining the mesh. In this sense, a criterion which relates the optimal mesh size to the adopted interface strength would be very useful and future work will address this issue. In the meantime it is desirable to have an algorithm which overcomes the limit points without falling into a 'black hole', as happens for the secant predictor strategy in the case of Figure 15(b), so that a preliminary analysis can be successfully completed even with a coarse mesh. Furthermore, sharp snap-backs can be the result of the actual structural behaviour, as will be shown later for a mixed-mode, multiple delamination.

The influence of the number of points used for evaluating the stiffness matrix and the residual force vector in the interface elements can be seen from Figure 16 which refers to a mesh of 4×100 plane strain elements enriched with enhanced strains (Q4M) and 70 linear interface elements (INT4) with a tensile strength of 32.0 MPa. Simpson rule has been used with 2 and 20 points and it is apparent that the increase of the number of integration points results in an increase

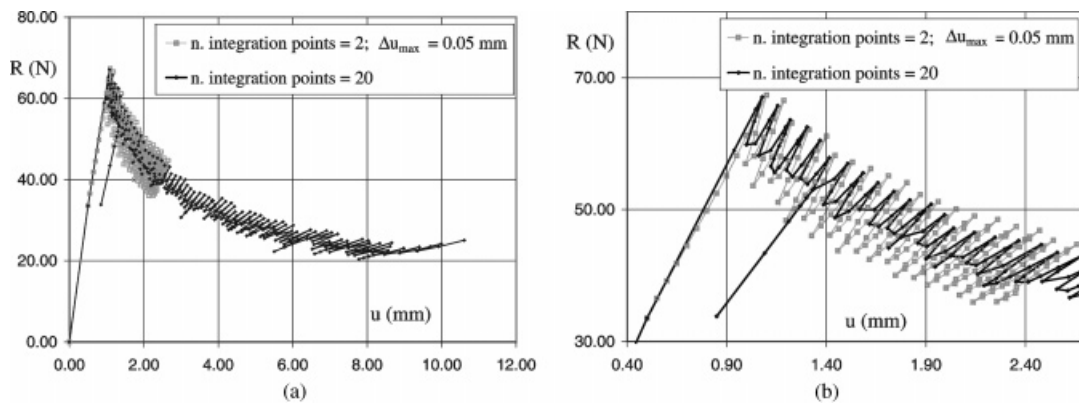


Figure 17. (a) 20 points vs 2 points with $u_{\max} = 0.05$ mm; (b) 20 points vs 2 points with $u_{\max} = 0.05$ mm (zoom).

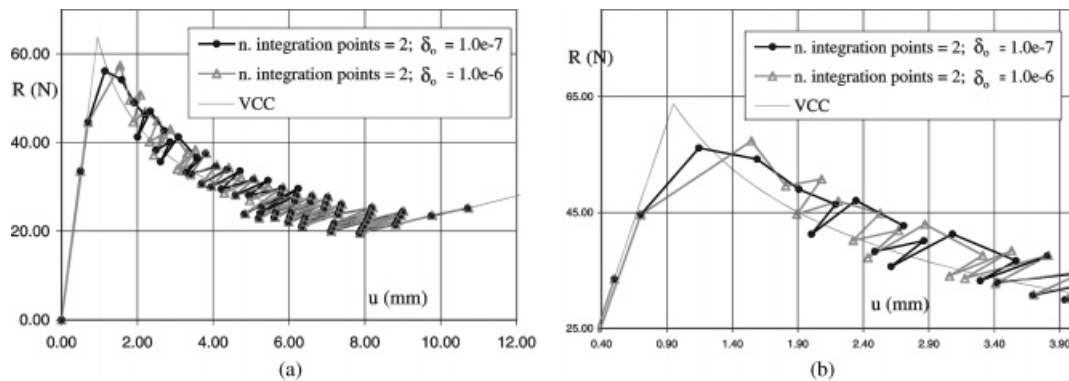


Figure 18. (a) Results with two integration points; (b) results with two integration points (zoom).

of the number of spurious oscillations of the reaction/displacement curve. In fact, for a higher number of integration points, the solution algorithm turns out to be less robust and the automatic incrementation procedure considerably reduces the step. Conversely, the higher stability achieved with two quadrature points allows for a larger average step size so that the solution often manages to 'jump over' several 'bumps'.

In Figure 17 a curve has been obtained using two integration points but fixing a maximum value $\Delta u_{\max} = 0.05$ mm for the change of the applied displacement in each increment. In this case the solution cannot 'jump over the bumps' and the result is similar to the one obtained with 20 points. This means that the frequency of the bumps in the real solution is indeed independent from the number of integration points, and that the reduced number obtained in Figure 16 for two points is related to the increased stability.

Furthermore, in order to investigate the general validity of the result, a small perturbation has been introduced by assigning, in the data of the interface, a different value of δ_o equal to 10^{-6} and 10^{-7} , respectively. The curves obtained, plotted in Figure 18, are different, but the general

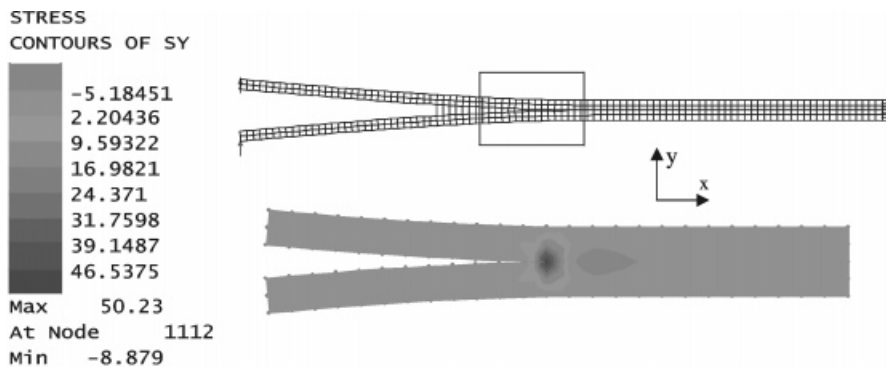


Figure 19. Contour-plot of σ_y for $t_o = 57.0$ MPa ($u_p = 2.66$ mm).

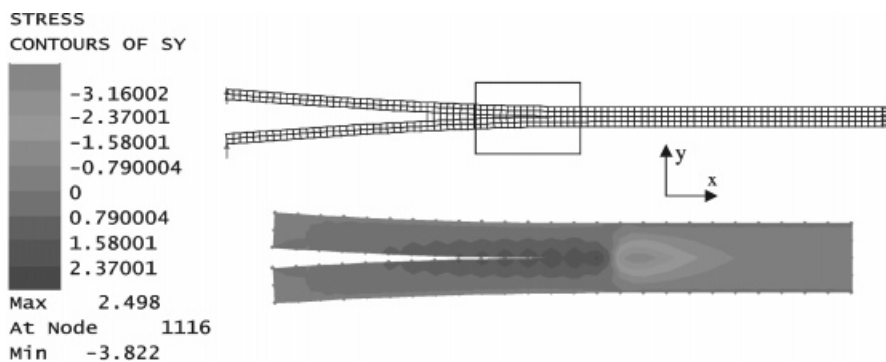


Figure 20. Contour-plot of σ_y for $t_o = 1.7$ MPa ($u_p = 2.55$ mm).

trend is similar and then the ability of 'jumping over the bumps' seems to be related with a higher stability rather than with pure luck. As a conclusion, increasing the number of integration points does not seem to be equivalent to a mesh refinement.

Figures 19 and 20 show the contour-plots of the component σ_y of the stress in the composite for a mesh of 4×400 Q4M and 280 INT4 elements and tensile strengths of 57.0 and 1.7 MPa, respectively.

They confirm that, in terms of the stress field, the solution is strongly dependent on the tensile strength, as one could expect. In contrast, the results in terms of load/displacement curves suggest that, for low values of t_o , a good agreement can be still obtained between the interface model and the LEFM formulation. However, this does not allow one to arbitrarily reduce the strength since, below a certain value, incorrect results can be obtained also in terms of the load/displacement curves. This is clearly shown for the more complex case of multiple delamination which will be addressed later on in this section.

Figure 21 shows a comparison between the load–displacement curves obtained with the use of linear 4-node interface elements (INT4) and Q4M parent elements and the ones resulting from using quadratic 6-node interface elements (INT6) with Q8 parent elements. Two different meshes

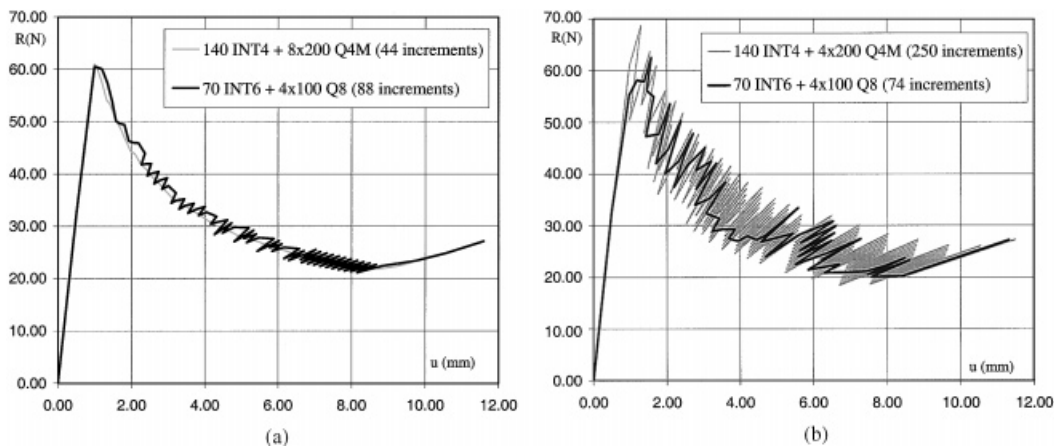


Figure 21. (a) Linear and quadratic elements: $t_o = 32.0$ MPa; (b) linear and quadratic elements: $t_o = 57.0$ MPa.

have been used which have a comparable number of nodes. For a lower value of the tensile strength, $t_o = 32.0$ MPa, the linear elements are more efficient, whereas in case of a higher tensile strength, $t_o = 57.0$ MPa, the performances of the quadratic elements are more satisfactory. This observation contrasts somewhat with the conclusion in Reference [21] that linear elements are more effective.

In this respect, it is worth noting that, when using the quadratic elements, the Simpson integration rule has been adopted to produce the results presented in this paper. Several numerical simulations revealed that the Newton–Cotes and the Simpson rule, with three points per element in both cases, can give different results in terms of stability and convergence rate, but it appears that no general rule can be deduced to predict in advance which one of them is the most convenient.

In order to compare the numerical results with an analytical solution [37], the same DCB has been studied considering an isotropic material with Young modulus $E = 2.1 \times 10^5$ MPa and Poisson ratio $\nu = 0.3$. A regular mesh of 4×400 four-node plane strain elements and 70 INT4 elements has been assumed.

Two analyses have been carried out using, for the first one, ordinary plane strain elements (Q4) and, for the second one, plane strain elements enriched with enhanced strains (Q4M). Figure 22 shows the importance of adding the enhanced modes to avoid shear-locking phenomena as well as a good agreement of the interface element model and the VCC method with the analytical solution.

In Figure 23 we show a comparison between the results of the interface damage model and some available experimental results provided in Reference [48]. In this case the composite is a XAS-913C, a carbon fibre epoxy composite whose material data are reported in Table II, and the geometry of the DCB specimen is the same as in Figure 11, except for the width which is now 30 mm. The agreement is good.

5.2. Mixed-mode multidelamination analysis

A case of multiple delamination has been analysed under displacement control. The geometry and the load conditions are illustrated in Figure 24 and the material data are given in Table III. They correspond to a (0_{24}) unidirectional HTA913 carbon-epoxy composite.

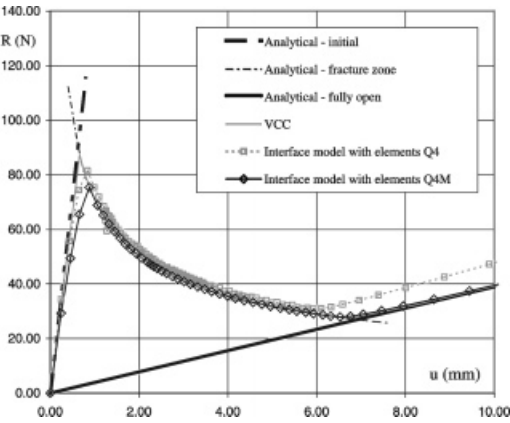


Figure 22. Isotropic material: comparison with an analytical solution.

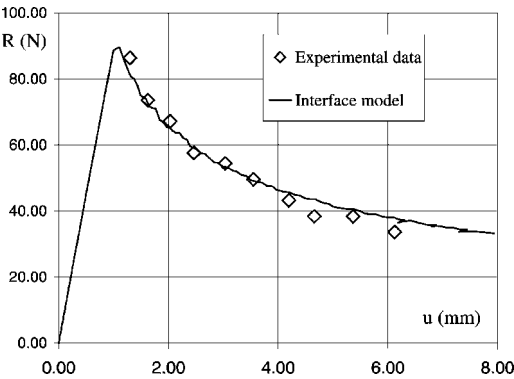


Figure 23. Comparison with experimental data reported in Reference [48].

Table II. DCB specimen: material data reported in Reference [48].

Composite XAS-913C			
$E_{11}(\text{GPa})$ 126	$E_{22} = E_{33}(\text{GPa})$ 7.5	$G_{12}(\text{GPa})$ 4.981	$\nu_{12} = \nu_{13} = \nu_{23}$ 0.263
Interface			
	$G_c(\text{N mm}^{-1})$ 0.281	δ_o 10^{-6}	$t_o(\text{MPa})$ 57.0

Table III. Multiple delamination analysis: material data.

Composite					
$E_{11}(\text{GPa})$ 115.0	$E_{22} = E_{33}(\text{GPa})$ 8.5	$G_{12}(\text{GPa})$ 4.5	$\nu_{12} = \nu_{13}$ 0.29	ν_{23} 0.3	
Interface					
$G_{c1}(\text{N mm}^{-1})$ 0.33	$G_{c2}(\text{N mm}^{-1})$ 0.80	$1 - \eta$ 1.0e−6	α 2	$t_{o1}(\text{MPa})$ 3.3	$t_{o2}(\text{MPa})$ 7.0

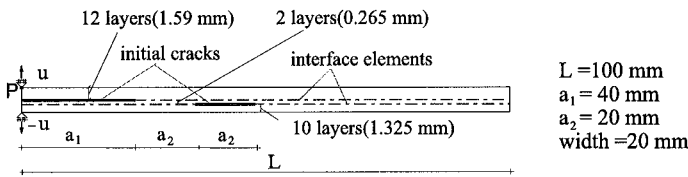


Figure 24. Multiple delamination analysis: geometry.

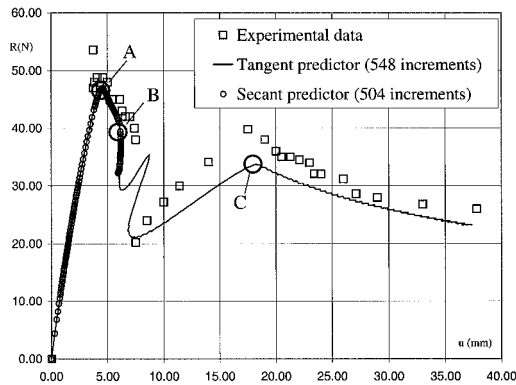


Figure 25. Reaction at P against prescribed displacement.

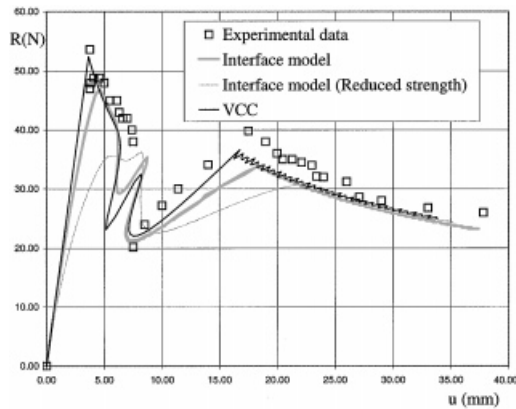


Figure 26. Comparison with the VCC method.

Two initial cracks are assigned. The one on the left is positioned at the middle plane of the specimen. The second one is positioned on the right of the first one, two layers beneath. The interface elements have been inserted at the not-initially delaminated parts of the interfaces between the 10th and the 11th layers and between the 12th and the 13th layers, starting from below.

A regular mesh of 3×360 Q4M elements and 680 INT4 interface elements has been used. In Figure 25 the reaction at the point P is plotted against the prescribed displacement. It shows a good agreement between the results of the interface model and the experimental data taken from Reference [8], as well as the failure of the secant predictor method in successfully completing the analysis.

The first part the curves of Figure 25 are almost linear with the slight non-linearity being due to the fact that some points immediately ahead of the upper initial crack are in the softening region although no one of them is completely delaminated before the point A has been reached.

Between points A and B the behaviour is approximately typical of a DCB specimen and, accordingly, only mode 1 is activated. Point B corresponds to a position of the crack tip which is approximately 10 mm before the second initial crack. The presence of the second initial crack now influences the delamination process, although only the upper crack propagates until point C in the curve is reached, which corresponds to a position of the upper crack approximately above the mid-point of the second initial crack. The numerical analysis predicts an extra local limit point that is not immediately obvious from the experimental results and might be an artefact of the mesh.

Starting from point C , the second initial crack propagates to the right and the two cracks grow simultaneously, the lower one moving approximately twice as fast as the upper one. A mixed-mode delamination occurs at this stage, although mode 1 is still more significant.

In Figure 26 a comparison is made between the results of the interface model and the curve obtained via the application of VCC. In this case of multiple delamination, for the VCC method, the energy release rates associated with the two cracks are evaluated at each step. Then the crack which is associated with the higher value of $G_1/G_{c1} + G_2/G_{c2}$ is moved, and the load multiplier is computed so as to fulfil the linear criterion (9). It is worth noting that the curve obtained by

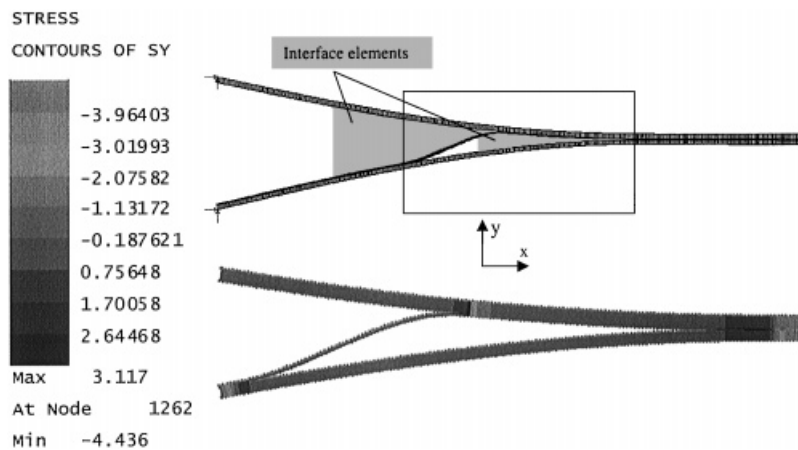


Figure 27. Multidelamination analysis: def. mesh and contour-plot of σ_y .

applying the VCC method shows a discontinuous trend in the part corresponding to the multiple delamination process. In fact, the simple implementation of the VCC method does not allow for a real simultaneous advance of both cracks so that, to circumvent this drawback, adaptive remeshing procedures should be used at each increment [8], with an increase of the computational burden.

Furthermore, in Figure 26 the curve obtained by reducing the strengths to $t_{o1} = 0.66$ MPa and $t_{o2} = 1.6$ MPa has been plotted. It shows that, if excessively low values of the strengths are assumed, the results can become unreliable.

In Figure 27 the deformed mesh and the contour-plot of the component σ_y in the composite are shown for a value $u = 37.2$ mm of the prescribed displacement.

6. CONCLUSIONS

The numerical analysis of the delamination in laminated composites has been addressed starting from an interface damage model originally presented in References [20–22]. The mixed-mode formulation has been modified to circumvent the inconsistency of reaching complete delamination in modes 1 and 2 at a different time. The model allows one to simulate the mixed-mode delamination without any hypotheses on the mode ratio so that no ‘pre-solution’ analyses are needed. This is because the interface relationship satisfies a generalized propagation criterion, introduced in References [20–22], which encompasses two criteria proposed in References [23, 24], often used in practical applications.

It has then been shown that the model can be recovered in the context of a damage mechanics theory as a regularization of a non-smooth model. The non-smooth model, in turn, reproduces the results of traditional fracture mechanics approaches when the penalty stiffness parameters, introduced to properly simulate the pre-crack behaviour, tend to infinity. In the framework of damage mechanics an alternative proof that the model satisfies the generalized mixed-mode criterion has been provided.

Some issues related with the numerical solution of the structural problem have been studied. The structural response is often characterized by sharp snap-throughs and snap-backs, which can either be due to the real behaviour of the structure or be a result of a finite element mesh not refined enough around the delamination front. In both cases it is important that the solution procedure be able to overcome these limit points and the strategy proposed in References [29, 30], as a modification of the quadratic arc length [27, 28], has proved to be the most robust so far, although further investigations on local control arc length procedures will be the object of future work.

The importance of using the tangent material stiffness in the predictor iteration at a point which has undergone delamination in the previous increment has been pointed out. This choice is consistent with an hypothesis of further delamination in the new increment and the numerical results presented show the improved rate of convergence and the enhanced numerical stability with respect to the case in which the secant material stiffness is used, which would be in accordance with an hypothesis of unloading.

Some of the numerical results presented have shown that increasing the number of integration points in the interface elements, in order to get a more accurate estimation of the tangent stiffness matrix and of the residual force vector, can reduce the stability of the solution algorithm. As a result, the undesired oscillations of the response for a coarse mesh are not eliminated whereas the number of required increments increases.

The sensitivity of the model to the interface strength has then been investigated. Reducing the values of the maximum admissible tractions lighten the computational burden because it allows for a coarser mesh around the crack tip and an increased average increment size. However, too small values can result in unacceptable errors and should be avoided.

The extension of the current treatment to the 3D case is quite simple and results will be presented in forthcoming papers. It can be accomplished by replacing the line interface element with a surface element and introducing mode III in the formulation, which is to be treated the same way as mode II although it is generally characterised by different material properties (in particular it may be that $G_{cII} \neq G_{cIII}$). In this respect it is worth noting that the 3D formulation can also be used for the analysis of delamination in plate and shell structures. Indeed many recent finite element formulations for shells discard rotations and use 'brick-like' elements in which the variables at the top and bottom nodes are assumed as unknowns.

APPENDIX A: DERIVATION OF THE INTERFACE MODEL FROM A DAMAGE MECHANICS THEORY: SINGLE-MODE DELAMINATION

Hereafter we show that the interface model described by Equation (4) can be recovered by using the damage mechanics formulation given by Equations (26)–(28) if relationship Equation (32) is assumed to hold true.

To this end, let us rewrite Equation (4) eliminating, for the sake of simplicity, any sign or component issues, analogously as in Section 3.1:

$$t = \begin{cases} K\delta & \text{if } \delta_{\max} \leq \delta_o \\ \left[1 - \left(\frac{\delta_{\max} - \delta_o}{\delta_{\max}} \right) \frac{\delta_c}{\delta_c - \delta_o} \right] K\delta & \text{if } \delta_o < \delta_{\max} < \delta_c \\ 0 & \text{if } \delta_{\max} \geq \delta_c \end{cases} \quad (A1)$$

Let us observe that

$$\delta_{\max} < \delta_o \Leftrightarrow \bar{Y} = \frac{1}{2} K \delta_{\max}^2 < G_o = \frac{1}{2} K \delta_o^2$$

Accordingly, if one initially sets $Y_c = G_o$ for $d = 0$, Khun–Tucker conditions (27) ensure that $\dot{d} = 0$ as long as $\delta_{\max} < \delta_o$, so that $(26)_1$ is equivalent to $(A1)_1$. For $\delta_{\max} = \delta_o$ we get $\bar{Y} = G_o$, which allows $\dot{d} \geq 0$. Then, for $d = 0$, $(26)_1$ is still equivalent to $(A1)_1$, whereas as soon as $d > 0$, $(32)_2$ has to be considered.

Since for $0 < d < 1$ we are in the softening part of the interface law, one can equate the expression of the traction in Equation $(26)_1$ to that given by Equation $(A1)_2$:

$$t = (1 - d)K\delta = \left[1 - \left(\frac{\delta_{\max} - \delta_o}{\delta_{\max}} \right) \frac{\delta_c}{\delta_c - \delta_o} \right] K\delta \quad (A2)$$

Hence, by making use of the relationships

$$\frac{\delta_c}{\delta_c - \delta_o} = \frac{G_c}{G_c - G_o} \quad \frac{\delta_o}{\delta_{\max}} = \frac{\sqrt{2G_o/K}}{\sqrt{2\bar{Y}/K}} = \sqrt{\frac{G_o}{\bar{Y}}} \quad (A3)$$

one gets

$$d = \frac{G_c}{G_c - G_o} \left(1 - \frac{\delta_o}{\delta_{\max}} \right) = \frac{G_c}{G_c - G_o} \left(1 - \sqrt{\frac{G_o}{\bar{Y}}} \right) \quad (A4)$$

and inverting

$$\bar{Y} = \frac{G_o}{(1 - ((G_c - G_o)/G_c)d)^2} \quad (A5)$$

Substituting the right-hand side of $(32)_2$ into (27), we get

$$Y \leq \frac{G_o}{(1 - ((G_c - G_o)/G_c)d)^2} \quad \dot{d} \geq 0 \quad \left(Y - \frac{G_o}{(1 - ((G_c - G_o)/G_c)d)^2} \right) \cdot \dot{d} = 0 \quad (A6)$$

It can be easily verified that, because of definition (30), conditions (A6) imply that (A5) holds true. Going backwards from (A5) to (A2) we then obtain that $(26)_1$ is equivalent to $(A1)_2$.

From (A5), in the limit for $d \rightarrow 1$, we get $\bar{Y} = G_c^2/G_o$ or, equivalently, $\delta_{\max} = \delta_c$. Accordingly, for $d = 1$, we get $\bar{Y} \geq G_c^2/G_o$, and then $\delta_{\max} \geq \delta_c$, whereas $(26)_1$ yields that $t = 0$, so that Equation $(A1)_3$ is recovered. Furthermore, being $\bar{Y} \geq G_c^2/G_o$, setting $Y_c \geq \bar{Y}$ makes condition $(27)_1$ automatically satisfied, whereas $(27)_{2-3}$ are trivially fulfilled because of (28).

However, for $d = 1$, as has been pointed out in Remark 3.2, being constantly $\dot{d} = 0$, Y , $\bar{Y}$ and Y_c do not have any real interest because the adhesion energy has already been completely released.

APPENDIX B: DERIVATION OF THE INTERFACE MODEL FROM A DAMAGE MECHANICS THEORY: MIXED-MODE DELAMINATION

The proof that the model described in Section 2.3 can be recovered by using the damage mechanics formulation of Section 3.2, if the threshold value ω_c is given by relationship (46), is conceptually similar to the one given in Appendix A for the one-dimensional model.

Let us first observe that, in accordance with Equation (37)₂, we have

$$\frac{\langle \delta_1 \rangle}{\delta_{o1}} = \frac{\sqrt{2Y_1/K_1}}{\sqrt{2G_{o1}/K_1}} = \sqrt{\frac{Y_1}{G_{o1}}} \quad \frac{\delta_2}{\delta_{o2}} = \frac{\sqrt{2Y_2/K_2}}{\sqrt{2G_{o2}/K_2}} = \sqrt{\frac{Y_2}{G_{o2}}} \quad (\text{B1})$$

Then the parameter γ , defined in Equation (14), can be expressed in terms of $\mathbf{Y}$ and the values G_{o1} and G_{o2} :

$$\gamma = \left[\left(\frac{Y_1}{G_{o1}} \right)^{\alpha/2} + \left(\frac{Y_2}{G_{o2}} \right)^{\alpha/2} \right]^{1/\alpha} - 1 \quad (\text{B2})$$

Introducing the regularization parameter of Equation (47) we get

$$\gamma = \frac{1}{\sqrt{1-\eta}} \left[\left(\frac{Y_1}{G_{c1}} \right)^{\alpha/2} + \left(\frac{Y_2}{G_{c2}} \right)^{\alpha/2} \right]^{1/\alpha} - 1 = \frac{f(\mathbf{Y})^{1/\alpha}}{\sqrt{1-\eta}} - 1 \quad (\text{B3})$$

and then

$$\bar{\gamma} = \max_{\tau' \leq \tau} \left\{ \frac{f[\mathbf{Y}(\tau')]}{\sqrt{1-\eta}} - 1 \right\} = \frac{\bar{\omega}^{1/\alpha}}{\sqrt{1-\eta}} - 1 \quad (\text{B4})$$

whereby

$$\frac{\bar{\gamma}}{\bar{\gamma} + 1} = 1 - \frac{\sqrt{1-\eta}}{\bar{\omega}^{1/\alpha}} \quad (\text{B5})$$

Setting $\omega_c = (1-\eta)^{\alpha/2}$ for $d = 0$, conditions (41) ensure that $\dot{d} = 0$ as long as $\bar{\gamma} < 0$, so that Equation (37)₁ is equivalent to (15)₁.

For $0 < d < 1$, if one equates expressions (15)₂ and (37)₂ and takes into account (B5), the following relationship is obtained:

$$d = \frac{1}{\eta} \left(1 - \frac{\sqrt{1-\eta}}{\bar{\omega}^{1/\alpha}} \right) \quad (\text{B6})$$

which, inverted, yields

$$\bar{\omega} = \left[\frac{1-\eta}{(1-\eta d)^2} \right]^{\alpha/2} \quad (\text{B7})$$

Substituting the right-hand-side of (46)₂ into (41) we get

$$f(\mathbf{Y}) \leq \left[\frac{1-\eta}{(1-\eta d)^2} \right]^{\alpha/2} \quad \dot{d} \geq 0 \quad \left\{ f(\mathbf{Y}) - \left[\frac{1-\eta}{(1-\eta d)^2} \right]^{\alpha/2} \right\} \dot{d} = 0 \quad (\text{B8})$$

which imply that (B7) holds true. Going backwards we can then conclude that (37)₂ is equivalent to (15)₂.

In the limit, for $d \rightarrow 1$, (B7) yields $\bar{\omega} = 1/(1-\eta)^{\alpha/2}$ or, equivalently (see also Equation (B5)):

$$\frac{1}{\eta} \left(\frac{\bar{\gamma}}{1+\bar{\gamma}} \right) = 1$$

Hence, for $d = 1$, $\bar{\omega} \geq 1/(1-\eta)^{\alpha/2}$ and $(1/\eta)(\bar{\gamma}/(1+\bar{\gamma})) \geq 1$, whereas Equation (37)₁ yields $\mathbf{t} = \mathbf{0}$, so that Equations (20) and (21) are recovered.

Furthermore, since it is $\bar{\omega} \geq (1-\eta)^{\alpha/2}$, setting $\omega_c = \bar{\omega}$ makes condition (41)₁ automatically satisfied whereas (41)₂₋₃ are fulfilled because of (28).

APPENDIX C: FULFILLMENT OF THE GENERALIZED ELLIPSE CRITERION: DAMAGE MECHANICS APPROACH

We here show that, if a proportional history of relative displacement is assigned, so that

$$\delta_2 = z\delta_1 \quad (\text{C1})$$

the GEC (11) is fulfilled.

Considering only the case $\delta_1 > 0$, let us observe that, using (C1), we can write

$$Y_2 = pY_1 \quad (\text{C2})$$

with $p = z^2 K_2/K_1$. Furthermore, we can rule out the case in which

$$\bar{\omega} > \left(\frac{Y_1}{G_{c1}} \right)^{\alpha/2} + \left(\frac{Y_2}{G_{c2}} \right)^{\alpha/2}$$

because it would yield $\dot{d} = 0$. Therefore we consider, without loss of generality, a monotonic loading so that

$$\bar{\omega} = f(\mathbf{Y}) = \left(\frac{Y_1}{G_{c1}} \right)^{\alpha/2} + \left(\frac{Y_2}{G_{c2}} \right)^{\alpha/2} \quad (\text{C3})$$

By differentiating Equation (B6), we then get

$$\frac{dd}{dY_i} = \frac{\sqrt{1-\eta}}{\eta} \frac{1}{2G_{ci}^{\alpha/2}} \frac{Y_i^{\alpha/2-1}}{f(\mathbf{Y})^{1/\alpha+1}} \quad (\text{C4})$$

and exploiting (C2) we can write

$$\begin{aligned} \int_0^{+\infty} Y_1 \dot{d}_1 dt &= \int_{Y'_1}^{Y''_1} Y_1 \frac{dd_1}{d\mathbf{Y}} \cdot d\mathbf{Y} \\ &= \frac{\sqrt{1-\eta}}{\eta} \int_{Y'_1}^{Y''_1} \frac{1}{2f(\mathbf{Y})^{1/\alpha+1}} \left[\left(\frac{Y_1}{G_{c1}} \right)^{\alpha/2} dY_1 + \left(\frac{Y_1 Y_2^{\alpha/2-1}}{G_{c2}^{\alpha/2}} \right) dY_2 \right] \\ &= \frac{\sqrt{1-\eta}}{\eta a} \int_{Y'_1}^{Y''_1} \frac{1}{2\sqrt{Y_1}} dY_1 = \frac{\sqrt{1-\eta}}{\eta a} [\sqrt{Y_1}]_{Y'_1}^{Y''_1} \end{aligned} \quad (\text{C5})$$

having set

$$a = \left[\left(\frac{1}{G_{c1}} \right)^{\alpha/2} + \left(\frac{p}{G_{c2}} \right)^{\alpha/2} \right]^{1/\alpha}$$

To obtain the integration limits Y_1' and Y_1'' let us write Equation (B7) using (C2) and (C3):

$$\begin{aligned}\bar{\omega} &= \left(\frac{Y_1}{G_{c1}}\right)^{\alpha/2} + \left(\frac{Y_2}{G_{c2}}\right)^{\alpha/2} = Y_1^{\alpha/2} \left[\left(\frac{1}{G_{c1}}\right)^{\alpha/2} + \left(\frac{p}{G_{c2}}\right)^{\alpha/2} \right] \\ &= Y_1^{\alpha/2} a^\alpha = \left[\frac{1-\eta}{(1-\eta d)^2} \right]^{\alpha/2}\end{aligned}\quad (C6)$$

Setting $d = 0$ and 1 , respectively, one gets

$$Y_1' = \frac{1-\eta}{a^2}, \quad Y_1'' = \frac{1}{(1-\eta)a^2} \quad (C7)$$

We then obtain

$$\frac{\int_0^{+\infty} Y_1 \dot{d}_1 dt}{G_{c1}} = \frac{\sqrt{1-\eta}}{\eta a G_{c1}} \left(\frac{1}{a\sqrt{1-\eta}} - \frac{\sqrt{1-\eta}}{a} \right) = \frac{1}{a^2 G_{c1}} \quad (C8)$$

Analogously we derive that

$$\frac{\int_0^{+\infty} Y_2 \dot{d}_2 dt}{G_{c2}} = \frac{p}{a^2 G_{c2}} \quad (C9)$$

so that we finally get

$$\left(\frac{\int_0^{+\infty} Y_1 \cdot \dot{d}_1 dt}{G_{c1}} \right)^{\alpha/2} + \left(\frac{\int_0^{+\infty} Y_2 \cdot \dot{d}_2 dt}{G_{c2}} \right)^{\alpha/2} = \frac{1}{a^\alpha} \left[\left(\frac{1}{G_{c1}}\right)^{\alpha/2} + \left(\frac{p}{G_{c1}}\right)^{\alpha/2} \right] = 1 \quad (C10)$$

APPENDIX D: FULFILLMENT OF THE GENERALIZED ELLIPSE CRITERION: DIRECT APPROACH

We report here an alternative proof of the result given in Appendix C, originally given in the internal report [37], which is based on the formulation presented in Section 2, allowing for different ratios δ_{oi}/δ_{ci} for the different modes and removing the hypothesis that $\delta_{o1} = \delta_{o2}$, assumed in Reference [37].

We want to prove that Equation (11) is fulfilled if the proportional relative displacement history (C1) is assumed, with z fixed during the delamination process, and the interface relationship of Section 2.2 is adopted. However, expressions (6) will now be used and substituted into (11).

Hence, let us consider expression (6)₁, assuming that $\delta_1 > 0$. From relationships (15) we can split the integral in two parts:

$$G_1 = \int_0^{\tilde{\delta}_1} K_1 \delta_1 d\delta_1 + \int_{\tilde{\delta}_1}^{\delta_1} \left(1 - \frac{\tilde{\gamma}}{\tilde{\gamma} + 1} F_{11} \right) K_1 \delta_1 d\delta_1 \quad (D1)$$

Without loss of generality we can assume a monotonic loading because, if unloading occurs, it would be followed by a loading path with a final zero balance of energy, whereby one can set $\bar{\gamma} = \gamma$ and write, using (C1):

$$\bar{\gamma} = \gamma = \left[\left(\frac{\delta_1}{\delta_{o1}} \right)^\alpha + \left(\frac{|\delta_2|}{\delta_{o2}} \right)^\alpha \right]^{1/\alpha} - 1 = \delta_1 \left[\left(\frac{1}{\delta_{o1}} \right)^\alpha + \left(\frac{|z|}{\delta_{o2}} \right)^\alpha \right]^{1/\alpha} - 1 = \delta_1 \beta_1 - 1 \quad (\text{D2})$$

having set

$$\beta_1 = \left[\left(\frac{1}{\delta_{o1}} \right)^\alpha + \left(\frac{|z|}{\delta_{o2}} \right)^\alpha \right]^{1/\alpha}. \quad (\text{D3})$$

The integration limits $\tilde{\delta}_1$ and $\hat{\delta}_1$ correspond to the beginning and to the end of the damage process, respectively. Setting $\bar{\gamma} = 0$ and $(\bar{\gamma}/(\bar{\gamma} + 1))F_{11} = 1$ we then get the two values:

$$\tilde{\delta}_1 = \frac{1}{\beta_1} \quad \hat{\delta}_1 = \frac{\kappa_1}{\beta_1} \quad \text{with} \quad \kappa_1 = \frac{F_{11}}{F_{11} - 1} \quad (\text{D4})$$

The evaluation of integral (D1) yields then, after some calculations

$$\begin{aligned} G_1 &= \int_0^{1/\beta_1} K_1 \delta_1 d\delta_1 + \int_{1/\beta_1}^{\kappa_1/\beta_1} \left[1 - \left(1 - \frac{1}{\delta_1 \beta_1} \right) F_{11} \right] K_1 \delta_1 d\delta_1 \\ &= \frac{K_1}{2\beta_1^2} \left(1 + \frac{\delta_{c1} - \delta_{o1}}{\delta_{o1}} \right) = \frac{K_1 \delta_{c1}}{2\beta_1^2 \delta_{o1}} \end{aligned} \quad (\text{D5})$$

Being $G_{c1} = K_1 \delta_{o1} \delta_{c1}/2$ we finally get

$$\frac{G_1}{G_{c1}} = \frac{1}{\beta_1^2 \delta_{o1}^2} \quad (\text{D6})$$

As for the second term in (6), we can write

$$\delta_2 = \frac{1}{z} \delta_1$$

so that we get an analogous relationship as (D6)

$$\frac{G_2}{G_{c2}} = \frac{1}{\beta_2^2 \delta_{o2}^2} \quad (\text{D7})$$

where

$$\beta_2 = \left[\left(\frac{1}{\delta_{o1}|z|} \right)^\alpha + \left(\frac{1}{\delta_{o2}} \right)^\alpha \right]^{1/\alpha} = \frac{1}{|z|} \left[\left(\frac{1}{\delta_{o1}} \right)^\alpha + \left(\frac{|z|}{\delta_{o2}} \right)^\alpha \right]^{1/\alpha} = \frac{\beta_1}{|z|} \quad (\text{D8})$$

Accordingly we get

$$\frac{G_2}{G_{c2}} = \frac{|z|^2}{\beta_1^2 \delta_{o2}^2} \quad (\text{D9})$$

and then

$$\left(\frac{G_1}{G_{c1}}\right)^{\alpha/2} + \left(\frac{G_2}{G_{c2}}\right)^{\alpha/2} = \left(\frac{1}{\beta_1^2 \delta_{o1}^2}\right)^{\alpha/2} + \left(\frac{|z|^2}{\beta_1^2 \delta_{o2}^2}\right)^{\alpha/2} = \frac{1}{\beta_1^\alpha} \left[\left(\frac{1}{\delta_{o1}}\right)^\alpha + \left(\frac{|z|}{\delta_{o2}}\right)^\alpha \right] = 1 \quad (\text{D10})$$

which is the desired result.

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